

Chapter 69

Using Viewers

This chapter covers working with viewers in Fusion, including using onscreen controls and toolbars, creating groups and subviews, managing viewer Lookup Tables (LUTs), working with the 3D viewer, and setting up viewer preferences and options.

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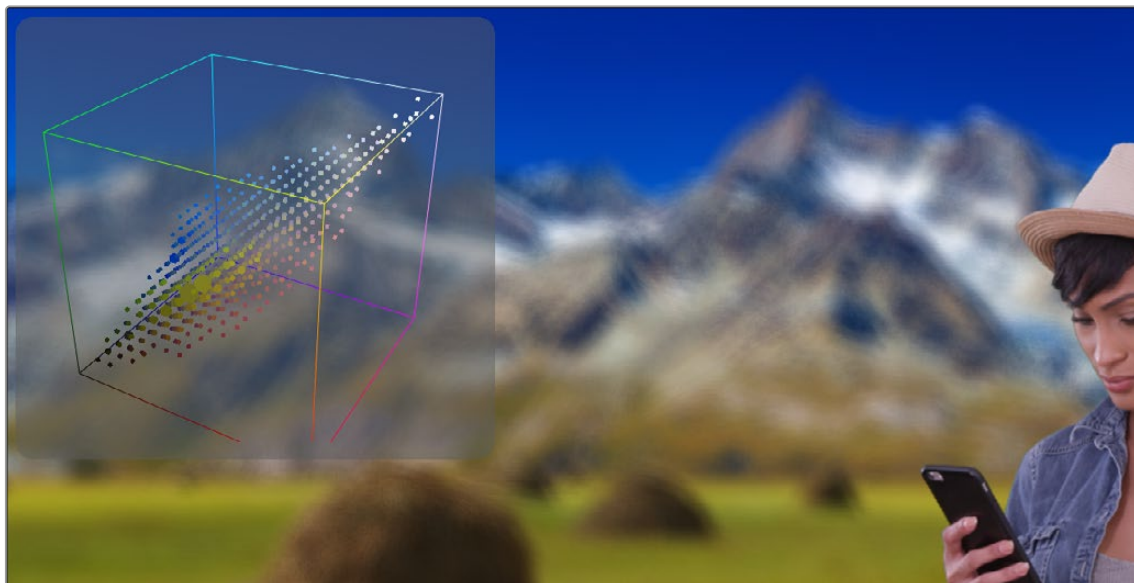
Viewer Overview

Viewers in Fusion display the current frame of the current composition in a variety of ways to help you see what you're doing and evaluate the final result of your compositing artistry. Viewers display 2D images, but they can also display a 3D environment using a 3D View as well as a special Quad viewer to help you effectively work in three dimensions.



Side-by-side dual viewers: a 3D viewer (left), and a 2D viewer (right)

Additionally, you can expose “subviews” including color inspectors, magnifiers, waveforms, histograms, and vectorscopes to help you analyze the image as you work.



Viewer with a 3D Histogram subview at the upper left-hand corner

Single vs. Dual Viewers

By default, there are two viewers positioned side by side across the top of the window. However, you can use the Single/Dual Viewer button to toggle between displaying a single viewer or two viewers displayed side by side.



The Single/Dual Viewer toggle button

Floating Viewers in Fusion Studio

In addition to the dual views above the Node Editor, Fusion Studio provides the option to use an unlimited number of floating viewers. These floating viewers are excellent for taking full advantage of a dual monitor configuration. Floating viewers can also be set to full-screen mode to make the best use of screen real estate.

To create a new floating display view, select Window > New View from the menu bar at the top of the screen. The position and configuration of the new view can be saved in the layout preferences, if required.

Video Output

When using DaVinci Resolve or Fusion Studio, if Blackmagic video hardware is present in the computer, then you can select a node to preview directly on that display. While video output can't be used for manipulating onscreen controls such as center crosshairs or spline control points, they're extremely valuable for evaluating your composition via the output format, and for determining image accuracy using a properly calibrated display.

The video hardware is configured from the DaVinci Resolve and Fusion Studio preferences.

Clean Feed

When using DaVinci Resolve with dual computer monitors, a full-screen viewer can be displayed on the secondary monitor from the Fusion page. This displays a third view indicator button under each node to control what is shown on the second display. To activate this monitor, make sure you do not have Dual Screen enabled under the Workspace menu and then select Workspace > Video Clean Feed and select your second computer display from the submenu.

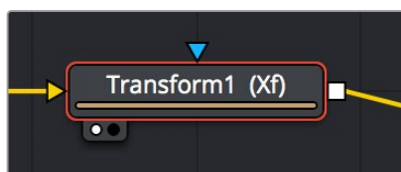
Loading Nodes into Viewers

The Fusion page in DaVinci Resolve and Fusion Studio show two different things when you first open each application. When you first open the Fusion page, the output of the current empty composition (the MediaOut1 node) is usually showing in viewer 2. If you're in dual-viewer mode, viewer 1 remains empty until you assign a node to one of them. In Fusion Studio, because there are no nodes when you first begin a comp, nothing is displayed in the viewers.

To load specific nodes into specific viewers, do one of the following:

- Hover the pointer over a node, and click one of two buttons that appear at the bottom-left of the node.
- Click once to select a node, and press 1 (for the left viewer) or 2 (for the right viewer).
- Right-click a node and choose View On > None/Left View/Right View in the contextual menu.
- Right-click the control header of a node in the Inspector, and choose View On > None/Left View/Right View from the contextual menu.
- Drag a node and drop it over the viewer you'd like to load it into (this is great for tablet users).

When a node is being viewed, a View Indicator button appears at the bottom left. This is the same control that appears when you hover the pointer over a node. Not only does this control let you know which nodes are loaded into which viewer, but they also expose little round buttons for changing which viewer they appear in.



Viewer assignment buttons at the bottom left of nodes indicate when they're being viewed, and which dot is highlighted indicates which viewer that node is loaded into.

Clearing Viewers

To clear an image from a viewer, click in the viewer to make it active; a light purple outline is displayed around the active panel. With the viewer active, press the ` (accent) key. This key is usually found to the left of the 1 key on U.S. keyboards. The fastest way to remove all images from all viewers is to make sure none of the viewers is the active panel, and then press the accent key.

Position and Layout

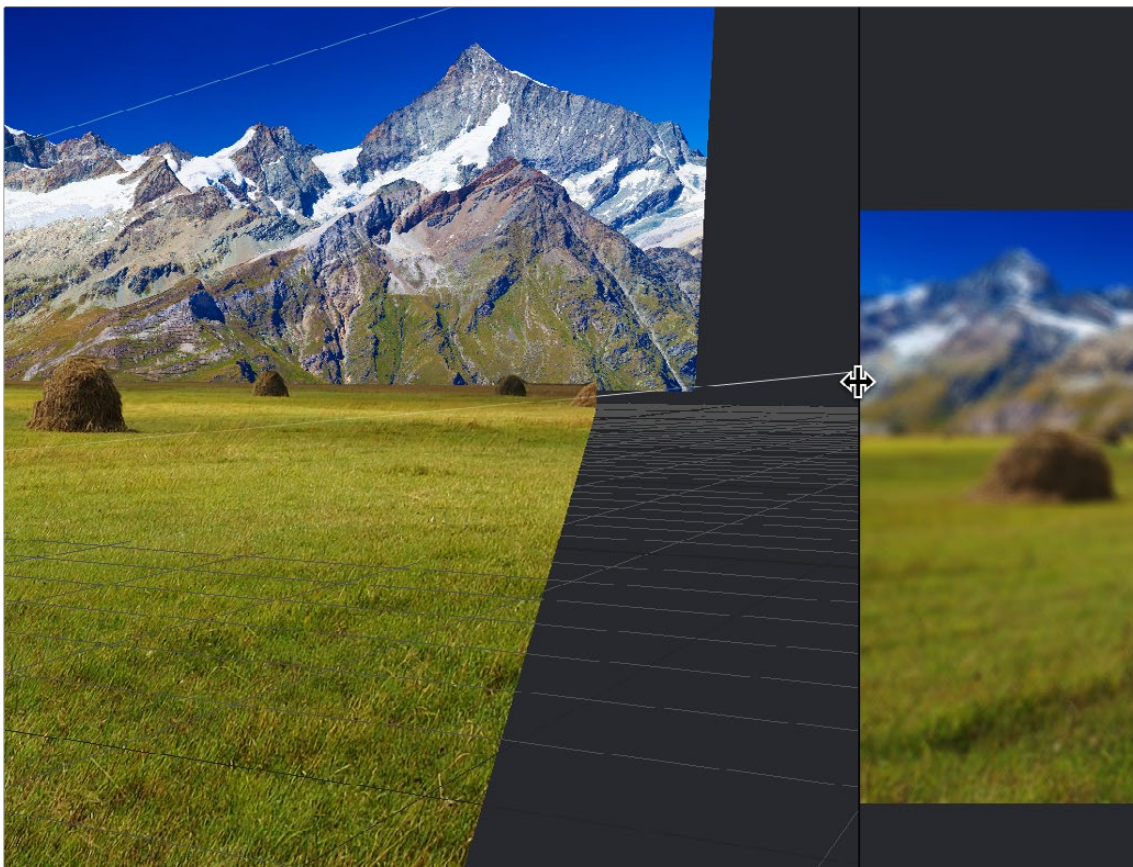
In Fusion Studio, when you resize and change the layout of viewers in the composition, that configuration is always saved with the composition. So each time you open the composition, the size and layout is remembered. You can prevent this behavior by disabling the Recall Layout checkbox in the Fusion Global Layout preferences.

If you want all new compositions to open with a certain viewer layout, you can configure the layout of the two primary viewers, and then use the Grab Document Layout button in the Fusion Global Layout preferences to remember the layout for any new compositions. To save the position and size of floating viewers, you use the Grab Program Layout button. Finally, if you want to have the floating viewers opened automatically when you open Fusion, enable the Create Floating Views checkbox.

The Viewer Divider

You can change the relative sizes of the left and right viewers using the horizontal viewer divider that runs between them. Drag the viewer divider to increase or decrease the amount of space used by one viewer. The adjacent viewer will adjust to accommodate the new layout.

The amount of vertical space available for both viewers can be adjusted by dragging the horizontal scrollbar between the viewers and the work area below them.



The viewer divider bar

Zooming and Panning into Viewers

There are standardized methods of zooming into and panning around viewers when you need a closer look at the situation. These methods also work with the Node Editor, Spline Editor, and Keyframes Editor.

Methods of panning viewers:

- Middle-click and drag to pan around the viewer.
- Hold down Shift and Command and drag the viewer to pan.
- Drag two fingers on a track pad to pan.

Methods of scaling viewers:

- Click a viewer and press the Equals key (=) to zoom in, and the Minus key (-) to zoom out.
- Press the middle and left mouse buttons simultaneously and drag left or right to resize the viewer.
- Hold down the Command key and use your pointer's scroll control to resize the viewer.
- Hold down the Command key and drag two fingers on a track pad to resize the viewer.
- Hold down the middle mouse button, and then click the left mouse button to zoom in, or click the right button to zoom out. The scaling uses a fixed amount, centered on the position of the cursor.
- Click a viewer and press Command-1 to resize the image in the viewer to 100 percent.
- Click a viewer and press Command-2 to resize the image in the viewer to 200 percent.
- Click a viewer and press Command-F or Command-1 to reset the image in the viewer to fit the viewer.
- Click the Scale Viewer menu and choose Fit or a percentage.
- Right-click on a viewer and choose an option from the Scale submenu of the contextual menu. This includes a Custom Scale command that lets you type your own scale percentage

Methods of spinning 3D viewers:

- In 3D Perspective view, hold down the middle mouse button and the right mouse button and drag to spin the stage around.

Flipbook Previews

As you build increasingly complex compositions, and you find yourself needing to preview specific branches of your node tree to get a sense of how various details you're working on are looking, you may find it useful to create targeted RAM previews at various levels of quality right in the viewer by creating a RAM Flipbook. RAM Flipbook Previews are preview renders that exist entirely within RAM and allow you to render a node's output at differing levels of quality for quick processing in order to watch a real-time preview.

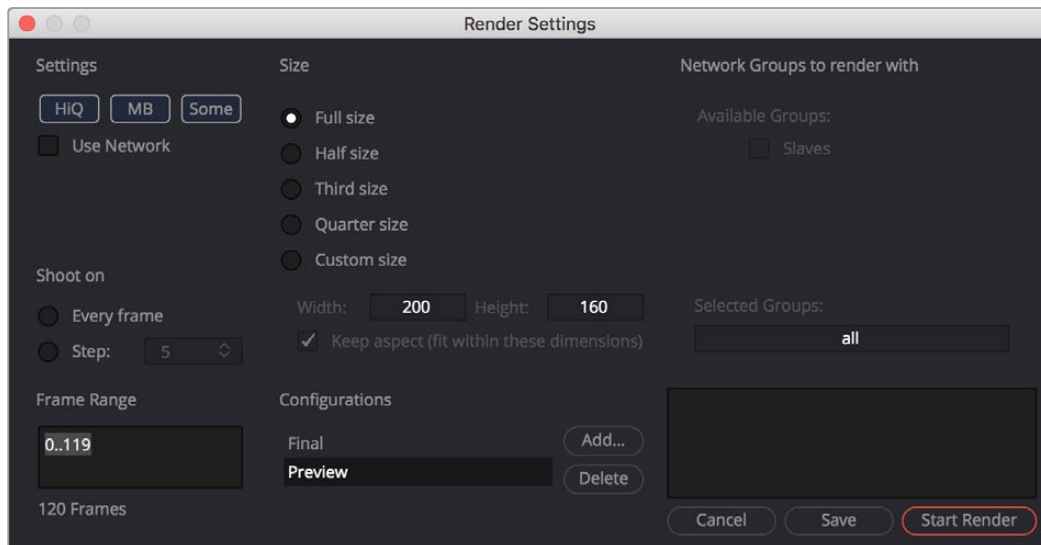
Creating Flipbook Previews

Creating a Flipbook Preview is relatively fast, once you know where to look.

To create a Flipbook Preview:

- 1 Choose the node in your node tree that you want to preview by doing one of the following:
 - Hold down the Option key while dragging a node into the viewer.
 - Right-click a node and choose an option from the Create/Play Preview submenu in the contextual menu.

- 2 When the Preview Render dialog opens, choose the quality, resolution, and motion blur settings you want to use for the Flipbook Preview.



The Flipbook Preview Render dialog

- 3 When you've chosen the settings you want to use, click Start Render.

The current frame range of the Time Ruler is rendered using the settings you've selected, and the result is viewable in the viewer you selected or dragged into.

Once you've created a Flipbook Preview within a particular viewer, right-clicking that viewer presents Flipbook-specific commands and options to Play, Loop, or Ping-Pong the Flipbook, to open it Full Screen, to Show Frame Numbers, and to eliminate it.

TIP: If you want to create a Flipbook Preview and bypass the Render Settings dialog by just using either the default setting or the settings that were chosen last, hold down Shift-Option while you drag a node into the viewer. The Settings dialog will not appear, and rendering the preview will start right away.

Playing Flipbook Previews

While the Flipbook Preview is loaded into a viewer, or open in full-screen mode, you can play or scrub through it using the mouse and the keyboard.

To play back a Flipbook using the mouse, do the following:

- Double-click in the viewer to start playback.

To scrub through a Flipbook using the mouse, do the following:

- Hold down the right mouse button down and drag left or right to scrub through frames.

To play back a Flipbook using the keyboard, do one of the following:

- Press the Spacebar to start or stop playback.
- Hold Shift and press the Spacebar to play in reverse.

To scrub through a Flipbook frame-by-frame using the keyboard, do one of the following:

- Press the Left or Right Arrow keys to move to the previous or next frame.
- Hold Shift and press the Left or Right Arrow keys to jump back or forward 10 frames.
- Press Command-Left Arrow to jump to the first frame.
- Press Command-Right Arrow to jump to the last frame.

TIP: The mouse and keyboard shortcuts work in full-screen mode as well.

Removing Flipbook Previews

Once you create a Flipbook Preview, you need to know how to clear it from RAM.

To eliminate a Flipbook you've created:

- Right-click within a viewer containing a Flipbook Preview, and choose Remove Preview.

Flipbook Preview Render Settings

This section covers all the settings available for rendering Flipbook Previews to RAM.

Settings

The Settings section of the Preview Render dialog includes three buttons that determine the overall quality and appearance of your Flipbook Preview. These buttons also have a significant impact on render times.

HiQ: When enabled, this setting renders the preview in full image quality. If you need to see what the final output of a node would look like, then you would enable the HiQ setting. If you are producing a rough preview to test animation, you can save yourself time by disabling this setting.

MB: The MB in this setting stands for Motion Blur. When enabled, this setting renders with motion blur applied if any node is set to produce motion blur. If you are generating a rough preview and you aren't concerned with the motion blur for animated elements, then you can save yourself time by disabling this setting.

Some: When Some is enabled, only the nodes specifically needed to produce the image of the node you're previewing are rendered.

Size

Since RAM Flipbook Previews use RAM, it's helpful to know how many frames you can render into RAM before you run out of memory. The Flipbook Preview dialog calculates the currently available memory and displays how many frames will fit into RAM. If you have a small amount of RAM in your computer and you cannot render the entire range of frames you want, you can choose to lower the resolution to a setting that delivers the best quality/duration ratio for your preview.

Network

Network rendering is only available in Fusion Studio.

For more information on network rendering, see *Chapter 66, "Rendering Using Saver Nodes,"* in the DaVinci Resolve Reference Manual or Chapter 4 in the Fusion Reference Manual.

Shoot On

Sometimes you may not want to render every single frame, but instead every second, third, or fourth frame to save render time and get faster feedback. You can use the Step parameter to determine the interval at which frames are rendered.

Frame Range

This field defaults to the current Render Range In/Out set in the Time Ruler to determine the start and end frames for rendering. You can modify the range to render more or fewer frames.

Configurations

Once you've created a useful preview configuration, you can save it for later use by clicking the Add button, giving it a name, and clicking OK.

Updating a Preview

This option is designed for the interactive frame-by-frame work of rotoscoping and painting. Right-click over a preview in the viewer and choose Update from its contextual menu. When active, any frames that are modified on the previewed node are automatically updated in the preview's playback. This lets you reserve the RAM for playback. You can keep it playing on a loop or ping-pong while you work in another viewer.

Onscreen Controls

When it comes to adjusting images, the Control Panel provides very precise numerical values, but sometimes visually positioning an element using onscreen controls can get you where you want to go with less tweaking.



The Angle preview control

The viewers show onscreen controls for manipulating the parameters of the currently selected node. Common onscreen controls include crosshairs, angle indicators, polylines, and paint strokes. Each of these controls can be manipulated directly in the viewer using the mouse or keyboard.

The controls shown in viewers are determined by which nodes are selected, not by the node displayed in the viewer. For example, a downstream blur is easily viewed while manipulating the controls for a selected polygon mask or merge. If multiple nodes are selected, the controls for every selected node are shown simultaneously.

Showing and Hiding Onscreen Controls

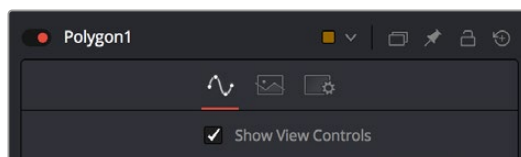
The onscreen controls for a viewer can be hidden so they don't interfere with viewing the image.

To toggle the visibility of onscreen controls, do one of the following:

- Click a viewer's Option menu and choose Show Controls to toggle the controls on or off.
- Right-click in a viewer and choose Options > Show Controls from the contextual menu.
- Select a viewer and press Command-K.

Enabling/Disabling Onscreen Controls in Specific Nodes

Some nodes, like masks, allow disabling of their onscreen controls on a per-node basis, since you often use multiple Polygon nodes to organize and animate masks.



You can disable some nodes, like the Polygon node, on a per-node basis.

Making Fine Adjustments to Onscreen Controls

If you want the visual guidance of onscreen controls with the precision of the Inspector, you can use different keyboard modifiers.

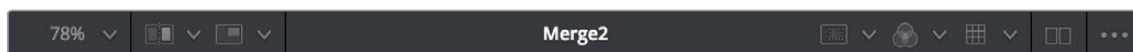
- Up and Down Arrow keys can be used to adjust the vertical position of an onscreen control by small steps.
- Holding down the Command key while using the Up and Down Arrow keys reduces the scale of each step by a factor of ten. Holding Shift increases the scale of each step by a factor of ten.

Toolbars

There are two toolbars in the viewer: a viewer toolbar, which always appears at the top of each viewer and gives you control over what that viewer shows, and an optional node toolbar that appears underneath that gives you contextual controls based on the node you've selected in the Node Editor.

Viewer Toolbar

A viewer toolbar runs across the top of each viewer, providing access to many of the most commonly used viewer-related settings, as well as an indication of the status of many of the most important settings. Most of the menus and buttons found on this toolbar are described in detail throughout this chapter.



The viewer toolbar

Node Toolbars

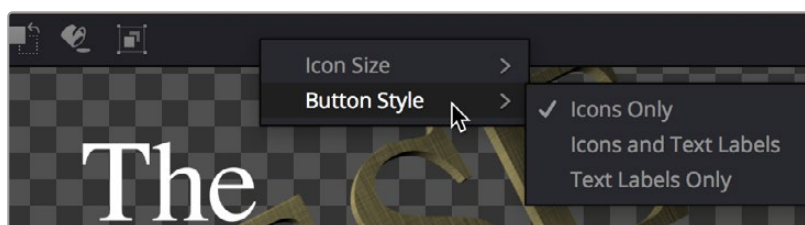
In addition to the viewer toolbar, a node toolbar is displayed underneath, at the top of the viewer display area, whenever you select a node that exposes special nodes. Examples of nodes that expose a toolbar include the text, masks, paths, paint strokes, and the 3D environment.



The node toolbar shown for the Paint node

Customizing the Node Toolbar

If you want to change the size of the buttons that appear in the Node toolbar, or turn on text names for each node, you can right-click anywhere in the empty area of the toolbar and choose new settings from the Icon Size and Button Style submenus in the contextual menu.



The contextual menu for the node toolbar

A/B Buffers

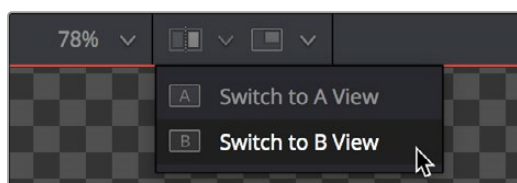
Each viewer has two buffers, each of which can contain images from different nodes, enabling easy comparison of two different nodes within the same viewer by either toggling between buffers, or via an adjustable split-wipe. Each buffer can be considered a complete and separate viewer within the same viewer pane. The A buffer is always shown by default, so when you first load a node into a viewer, the image loads into the A buffer.

Flipping between Buffers

Switching between buffers is easy, either to view a different image while keeping another image handy, or to flip between the original image and the affected image for comparison.

To switch between buffers, do one of the following:

- Select a viewer and press comma (,) to select the A buffer or press period (.) to select the B buffer.
- Click the Buffer menu and choose either Switch to A View or Switch to B View.



The Buffer menu lets you switch between buffers

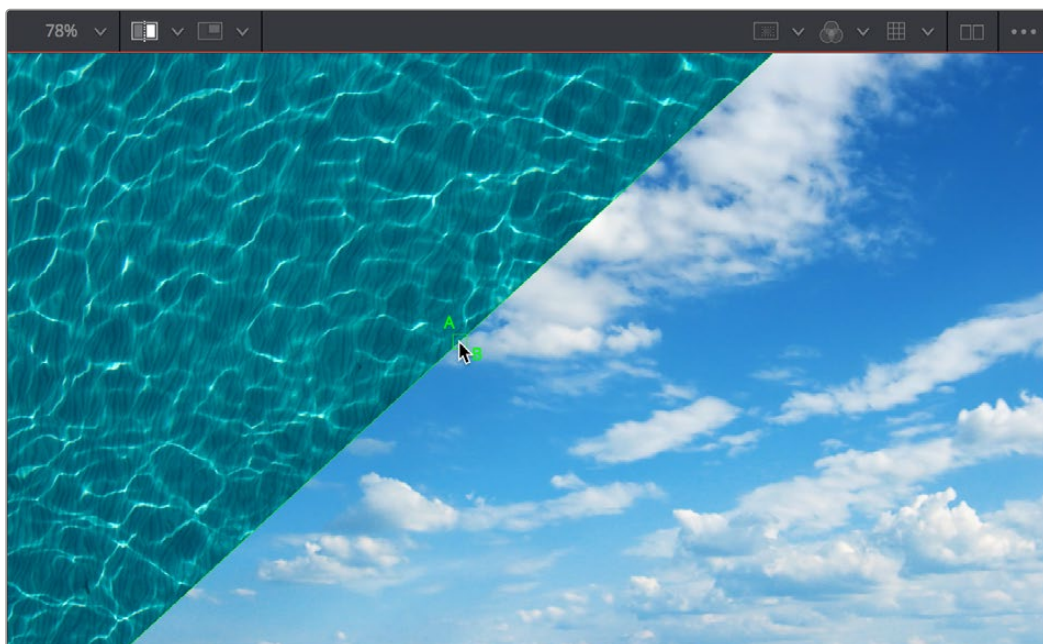
TIP: Each buffer can be set to different display settings—for example, showing different channels or different viewing LUTs, either applied to different nodes or applied to two buffered versions of the same node.

Split Wipes between Buffers

You can also wipe between both buffers, providing a more direct means of comparison.

To wipe between buffers, do one of the following:

- 1 Prepare to wipe between two images by loading different nodes into each buffer, or load the same node with different viewer options into each buffer.
- 2 To toggle the split wipe on or off, do one of the following:
 - a) Click the Switch to Split Wipe View button.
 - b) Press Forward Slash (/).
- 3 To adjust the wipe, do one of the following:
 - a) Move the center of the wipe by dragging the center handle of the wipe divider.
 - b) Press Command-Option and click anywhere in the viewer to jump the wipe divider to that location.
 - c) Change the angle or the wipe by dragging the wipe divider. Dragging the wipe divider while holding the Shift key snaps it to the nearest 45-degree angle.
 - d) Panning or zooming the viewer pans and zooms both buffers together.
- 4 (Optional) If you want to change the image that's displayed on that side of the split, you can drag new nodes onto either half of the viewer.
- 5 To turn off the wipe, click the Switch to Split Wipe View button again (or press /).



The wipe divider can be adjusted for comparing different areas of the A and B images

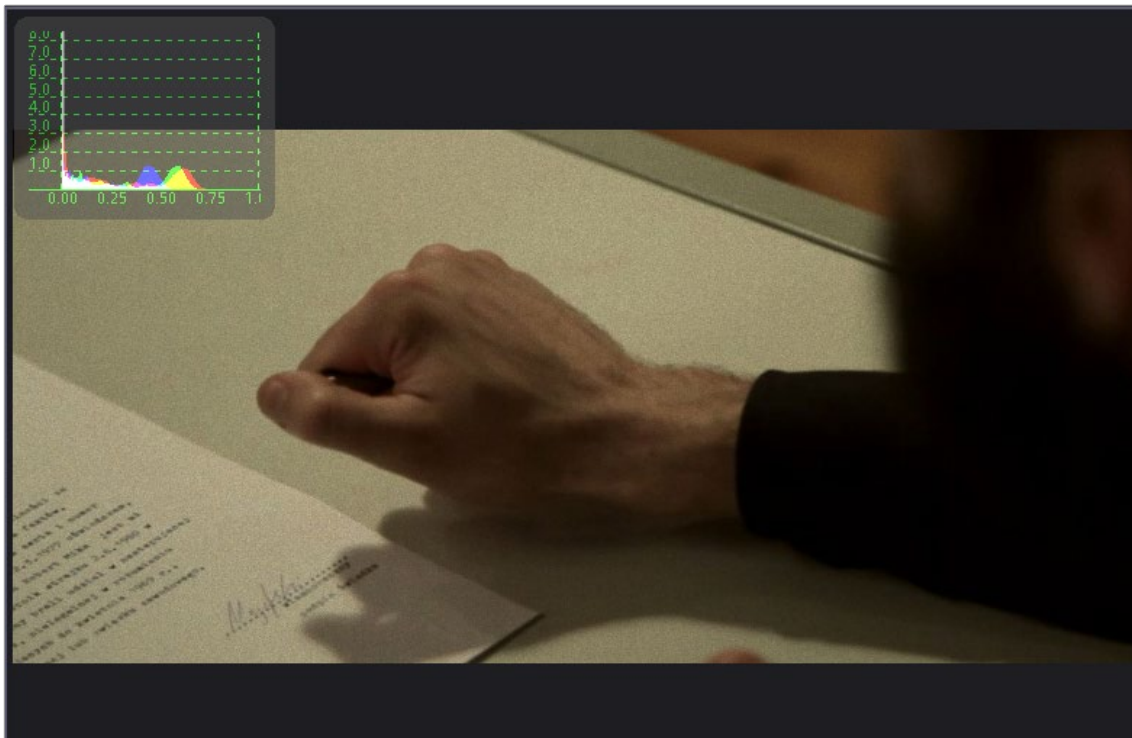
Even when you wipe, you can choose different display channels, view LUTs, or other display options for each buffer individually by clicking on the half of the wipe you want to alter, and then choosing the options you want that buffer to use. This allows easy comparison of different channels, LUTs, or other viewer settings while wiping the same image, or different images.

Moving the Wipe Divider

Occasionally, you will have either zoomed in or panned so far from the viewer divider that it's no longer visible in the viewer. Holding down Command-Option and clicking anywhere in the image will cause the viewer divider to jump to the current position of the pointer.

Subviews

A subview is a “mini” viewer that appears within the main viewer. A subview is usually used to show different information about the image.



The Subview menu with the Histogram subview displayed

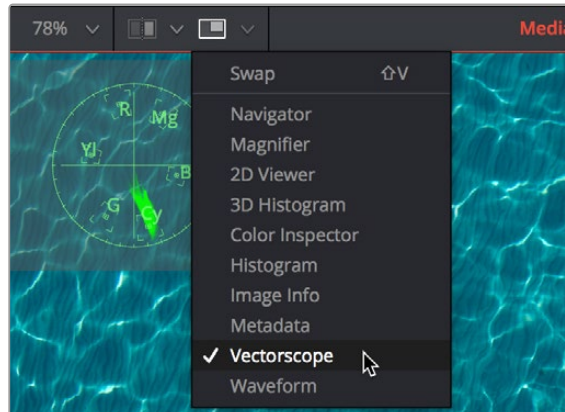
For example, the RGB channels can be viewed in the main viewer, while the alpha channel is displayed in a subview. For the most part, the subview is a fully functional miniature viewer, with its own contextual menu and options. It responds to the same keyboard shortcuts and navigation controls as any other viewer. However, there are several view types designed for use only in the subview, including the Navigator, Magnifier, Color Inspector, and Image Info.

Showing and Hiding Subviews

Subviews are easily shown and hidden.

To enable the currently selected subview in the Subview menu of a viewer, do one of the following:

- Click the Subview button in the View toolbar.
- Choose Views > Subview > Enabled from the contextual menu.
- Click a viewer, and press the V key.



The Subview button in the viewer toolbar

Changing the Subview Type

The Subview button enables and disables the subview, which usually shows the last subview you chose. You can change this at any time.

To change which subview type is displayed, do one of the following:

- Click the small arrow to the right of the Subview button to open its menu and choose which subview you want.
- Right-click within a subview to bring up the subview's contextual menu.

The Subview drop-down menu and contextual menu show all the available subview types. Once you choose an option from the list, that view will be displayed in the subview, and the Subview button will show and hide it as you wish.

Swapping the Subview with the Main View

It's possible to swap the contents of the main viewer and the subview for select subview types. However, certain view types, such as the Color Inspector and Magnifier, can only be used in the subview. In these cases, the swap will not take place.

To swap the contents of the subview with the main view, do one of the following:

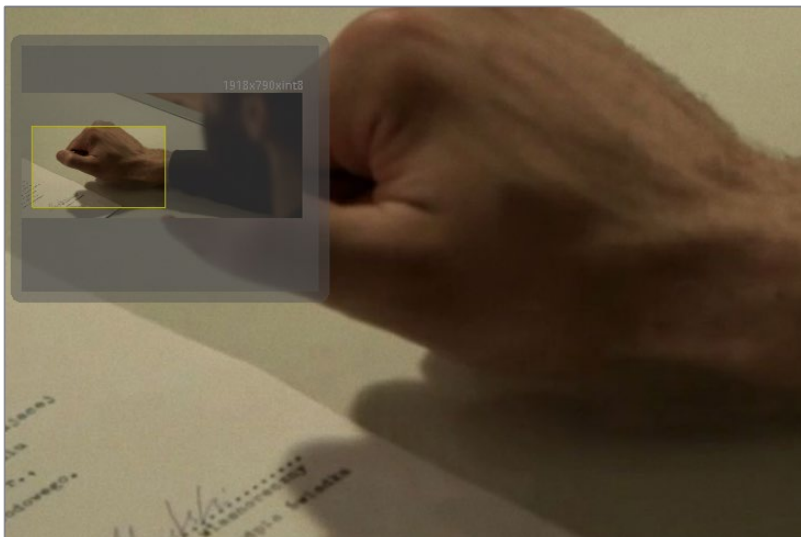
- Press Shift-V.
- Right-click in a viewer and choose Views > SubView > Swap from the contextual menu.

Viewer and Subview Types

Viewers can be changed to show a variety of different information about the image, but not all view types are available at all times. For example, the 3D Viewer is not available for a 2D node, and some of the measurement viewers are available only as subviews. Below is detailed information about the different view types available.

Navigator

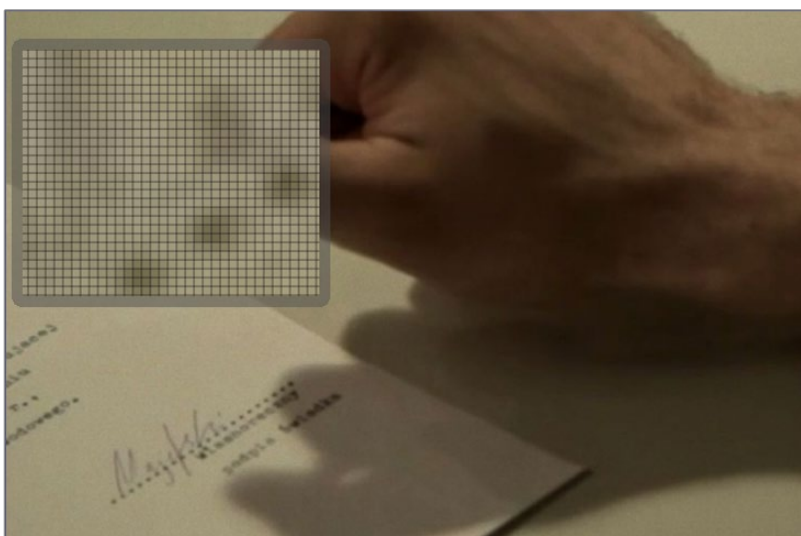
The Navigator can only be used in a subview. It provides a small overview of the entire image, with a rectangle that indicates the portion of the image that is actually visible in the main viewer. This is useful when zooming in on an image in the main view.



The Navigator subview for panning the image while zoomed in

Magnifier

The Magnifier can be used only in a subview. It shows a zoomed-in version of the pixels under the cursor in the main viewer.

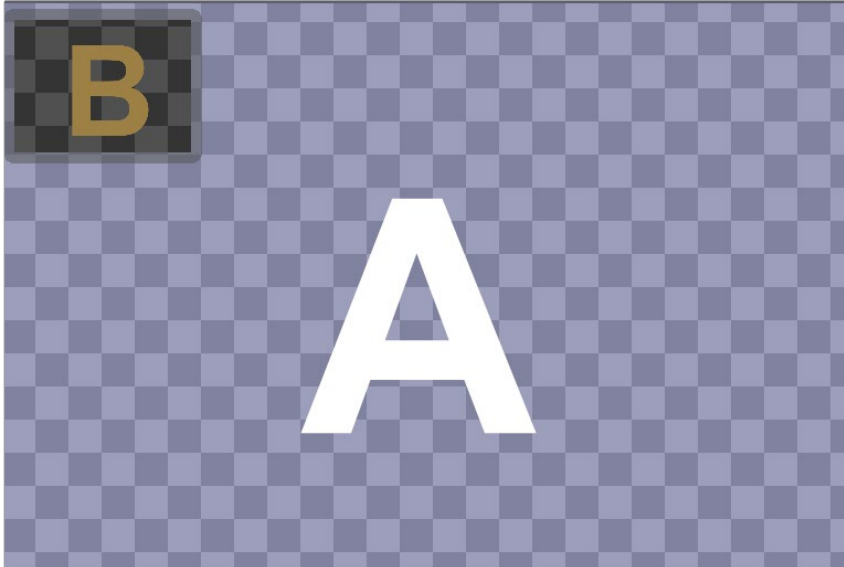


The Magnifier subview used to view a zoomed-in version of the image

2D Viewer

The 2D Viewer is the default type for showing images. When used as a subview, a different node than the one used in the main viewer can be displayed by dragging the node into the subview.

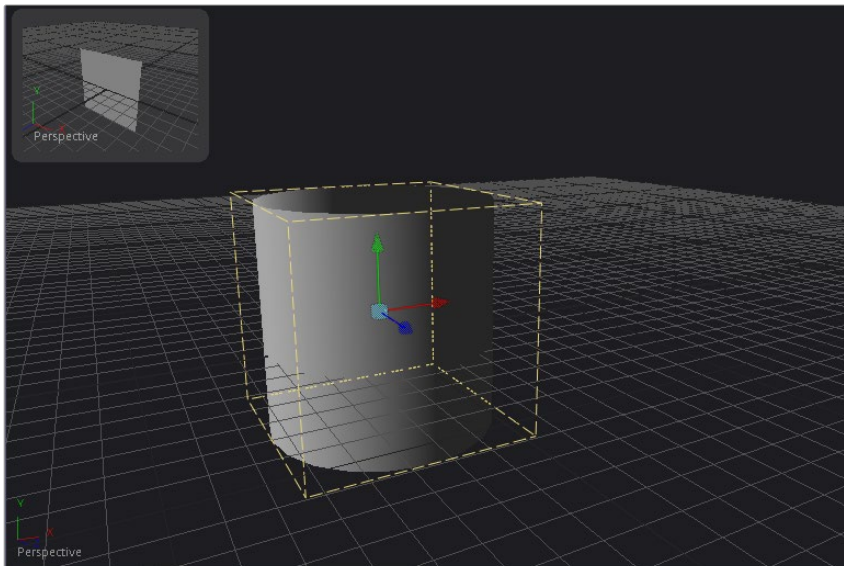
This is the only subview type that is not just a different view of the same node in the main viewer.



The subview used as another viewer

3D Image Viewer

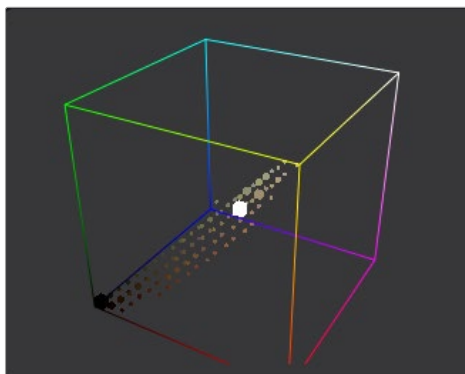
The 3D Image Viewer is available when viewing a node from the 3D category.



A 3D Image Viewer as a subview

3D Histogram

The more advanced 3D Histogram Viewer shows the color distribution in an image within a 3D cube. One advantage to a 3D Histogram is that it can accurately represent the out-of-range colors commonly found in floating-point and high-dynamic-range images. It can also be used to look at vector images like position, normal, velocity, and so on.



The 3D Histogram Viewer type for evaluating out-of-range colors

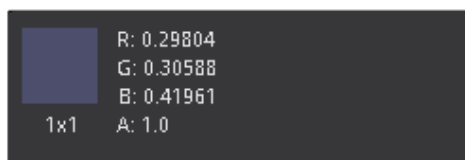
To rotate within a 3D Histogram, do one of the following:

Hold down the Option key, and drag left or right using the middle mouse button.

Hold down the middle and right mouse buttons while dragging.

Color Inspector

The Color Inspector can only be used in a subview. The Color Inspector shows information about the color channels of the pixel under the cursor. It will show all channels present, even the auxiliary channels such as Z buffer, XYZ normals, and UV mapping channels.

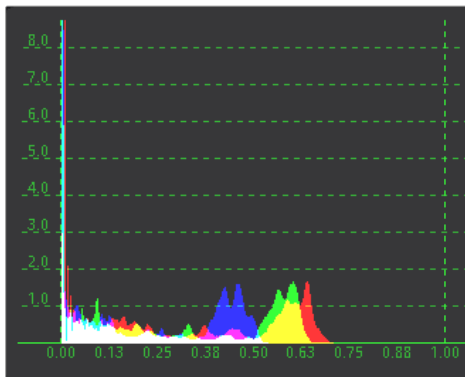


The Color Inspector subview for evaluating colors in an image

Histogram

The Histogram Viewer is an analysis node that can be used to identify problems with the contrast and dynamic range in an image. The graph shows the frequency distribution of colors in the image, including out-of-range colors in floating-point images. The horizontal axis shows the colors from shadows to highlights. The vertical axis shows the number of pixels in the image that occur at each level.

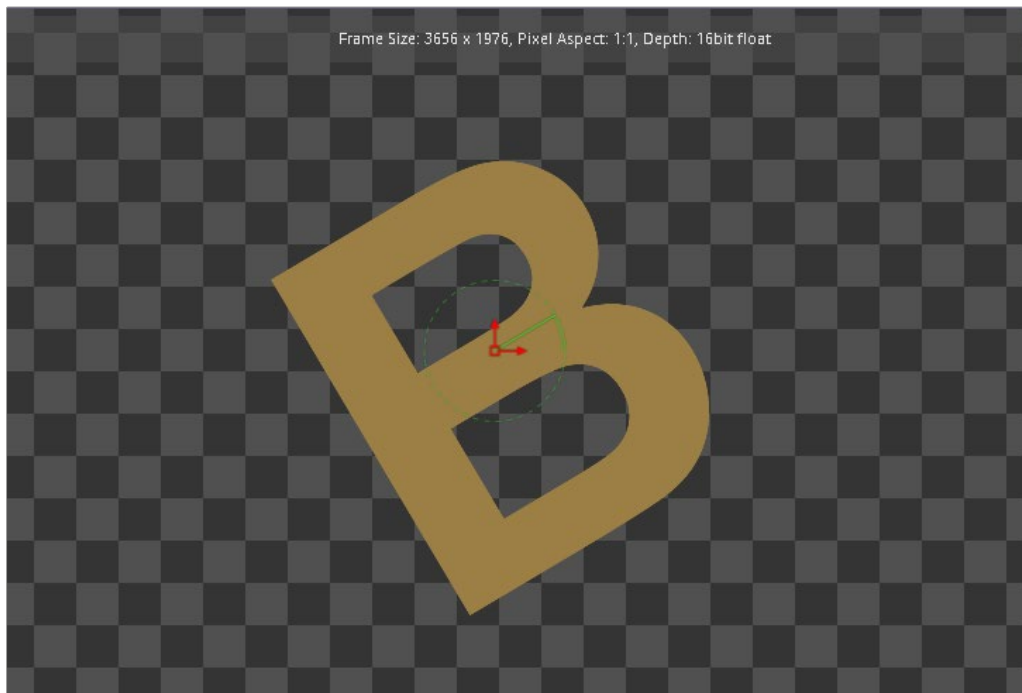
The Histogram Viewer will also display gradient information. You can use the From Image and Perturb modifiers to output gradients. If you need to see the gradient represented in a histogram, drag the modifier's title bar into the viewer.



The Histogram Viewer type for evaluating the contrast and color cast in an image

Image Info

The Image Info view can only be used in a subview. The Image Info tab shows a horizontal bar across the top of the image with information about the frame size, pixel aspect, and color depth of the viewed image.



The Image Info subview for viewing size, pixel aspect, and color depth information

Metadata

The content of this subview is based entirely on the amount of metadata in your image. Most Loaders will give the color space and file path for the image. Much more information can be displayed if it exists in the image.

```
CreationTime = 2013:10:11:14:51:36
screenWindowCenter = { 0, 0 }
screenWindowWidth = 1
Filename = C:\temp\out.exr
```

The Metadata subview for viewing embedded metadata

Vectorscope

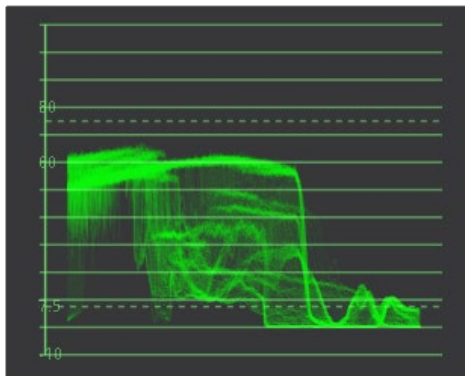
The Vectorscope Viewer duplicates the behavior of a specific type of video test equipment, displaying a circular graph that helps to visualize the intensity of chrominance signals.



The Vectorscope Viewer type for evaluating chrominance in an image

Waveform

The Waveform Viewer duplicates the behavior of a specific type of video test equipment, displaying a line or bar graph that helps to visualize the voltage or luminance of a broadcast signal.



The Waveform Viewer type for evaluating luminance in an image

Viewing Selective Channels

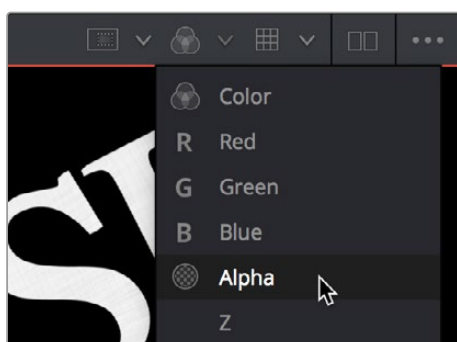
When compositing, you often deal with individual color components or channels in an image as much as you deal with the full RGB color of the entire image. The viewers and subviews can display the isolated color, alpha, depth channels, and even auxiliary channels that make up the image.

Viewing Color Channels

The default view is the full RGB color channel, but to change the channel that is displayed you can use the Channel toolbar button, the viewer's contextual menu, or keyboard shortcuts.

To toggle between RGB and alpha channels in the active viewer:

- Click the Color button in the viewer toolbar to toggle between full RGB color and that image's alpha channel.



The Channel toolbar button for switching between color, alpha, and depth channels

To toggle the channel that's displayed in the active viewer:

- Click the arrow to the right of the Color button to choose a specific channel to view from the list of available channels in the current image.
- Click the viewer you want to toggle, and press one of the following keyboard shortcuts;
 - C - Full RGB color display
 - R - Display red channel
 - G - Display green channel
 - B - Display blue channel
 - A - Display alpha channel
 - Z - Display Z-buffer channel

Viewing Auxiliary Channels

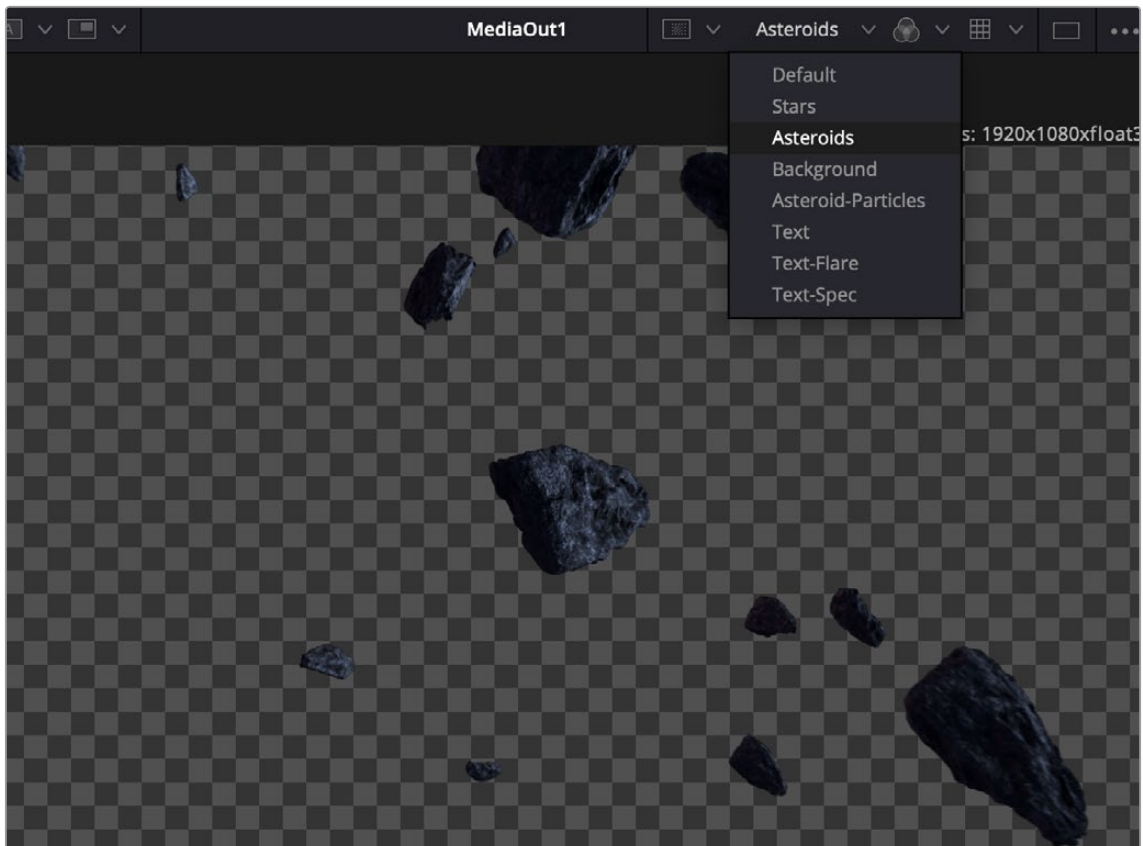
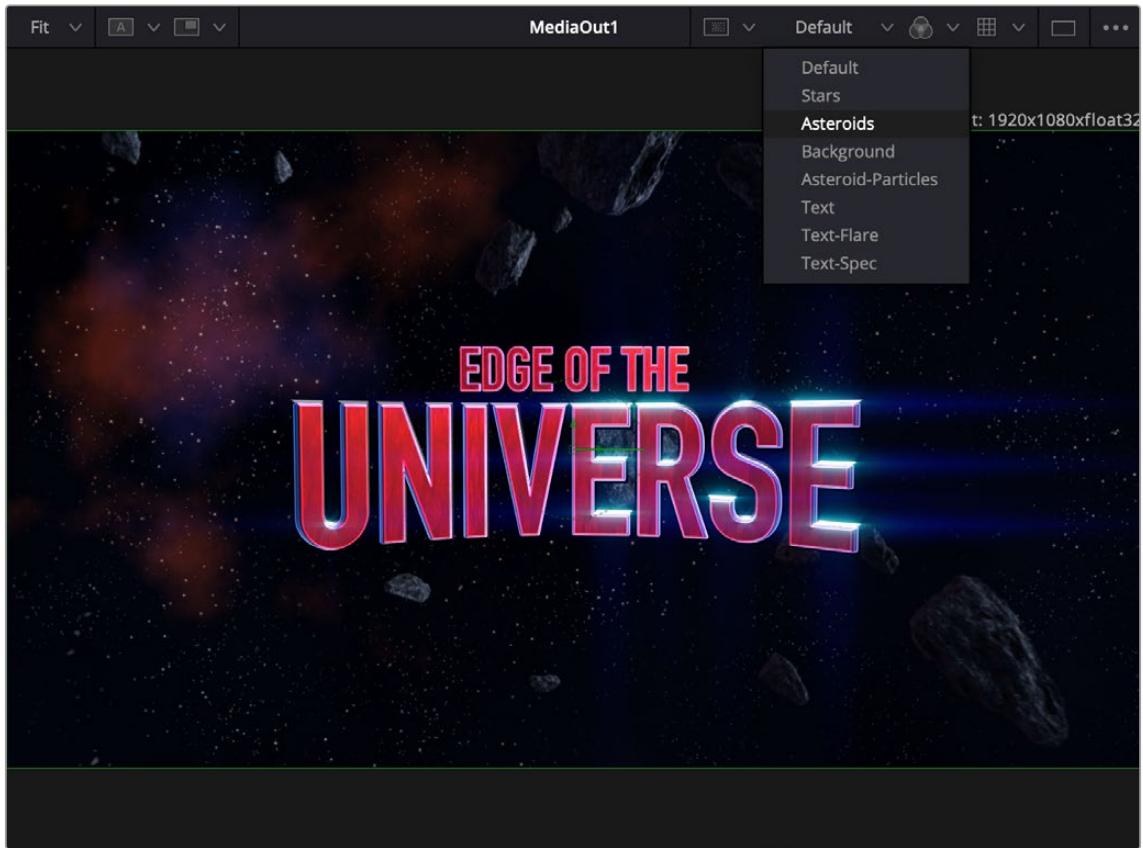
The viewers support RGBA and Z channels using keyboard shortcuts, but they support other channels as well. File formats such as OpenEXR often include auxiliary image data that provide more control and compositing options when working with rendered 3D images. To view auxiliary image data in a viewer, click the arrow to the right of the RGB button to display the drop-down menu or right-click in the viewer and choose an option from the Channels submenu of the contextual menu.

Viewing Multilayer Files (Studio Version Only)

Fusion can intelligently handle multilayer image workflows with support for EXR, Stereoscopic 3D, and PSD assets. Users can see a list of layers on the Fusion viewers and interact with layers in each node in the Settings tab in the Inspector. As this layer structure is retained throughout the toolset, this means you can isolate a tool to work on an individual layer, while still passing through the other layers unmodified.

Viewer

When loading a multilayer file, you can preview specific layers in the Viewer and selecting a layer from the drop-down list. Files without layer support will show only a 'default' option from the drop-down menu.



Using the Layer drop-down in the Viewer to select just the Asteroids layer from a multilayer file for viewing

The 3D Viewer

Building a composite in 3D space has different requirements from traditional 2D compositing. When a node from the 3D category or some particle systems is selected, a 3D Viewer is used to display the scene. The 3D Viewer shows a representation of a composite in a true GPU-accelerated 3D environment.

For more information on 3D controls, see *Chapter 85, “3D Compositing Basics,”* in the DaVinci Resolve Reference Manual or Chapter 25 in the Fusion Reference Manual.

Panning, Scaling, and Rotating a 3D Viewer

For the most part, navigation in the 3D Viewer is similar to the navigation in the 2D Viewer. Panning and zooming work with the same controls even though you’re moving within a 3D space. However, when viewing a 3D scene, panning changes the point of view and thus the center point for scaling and rotation, too. A combination of panning and rotation will allow you to move the point of view anywhere in the scene.

Another small change is that there’s a lower limit to the scale of a 3D scene. Continuing to zoom in past this limit will instead move (“dolly”) the point of view forward. The mouse wheel will move forward slowly, and the keyboard will move more quickly.

Critically, the 3D Viewer gives you additional control to rotate the viewer within the three dimensions of the scene to better see your scene from different angles as you work.

To rotate within a 3D Viewer, do one of the following:

- Hold down the Option key and drag left or right using the middle mouse button.
- Hold down the middle and right mouse buttons while dragging.

The rotation is centered on the middle of the view.

TIP: These rotation controls can be used with the 3D Histogram subview as well.

Viewing Objects via Wireframe

3D composites not only work with 2D images on image planes but can also integrate true geometry, such as that generated by the Particle system, Text 3D node, imported FBX meshes, and basic primitives from the 3D toolset. Using a Wireframe view helps to see through a mesh or see the density of the geometry. It is much easier to see a change in the Subdivision level of an ImagePlane3D in wireframe than viewing the rendered image.

To display 3D geometry in wireframe, do the following:

- Right-click the 3D Viewer and choose 3D Options > Wireframe from the contextual menu.

Changing the POV of a 3D Viewer

Compositing a 3D scene often requires that you view the scene from different angles to make sure the position and animation are what you want. While the 3D Viewer uses a perspective camera that allows you to look at the 3D scene from any angle, you can also switch the 3D Viewer to view from the front, top, left, or right side of the scene; these are collectively called Orthographic views.

Additionally, if you have a camera or spotlight in your scene, you can switch the viewer to face the scene from the point of view of those objects.

To change the 3D viewpoints:

Right-click the viewer and choose an option from the Camera submenu of the contextual menu. The choices include Perspective, Front, Top, Left, and Right.

Changing Cameras in a 3D Viewer

If you have one or more camera objects in the 3D scene, they will be listed as options in the contextual menu. Spotlights and other lights or objects in the scene will be listed in the Other submenu. If you choose any one of these objects, the 3D Viewer jumps to display the scene from the point of view of the chosen object. While looking “through” an object, rotating, panning, or zooming, the viewer will instead change the rotation, position, and scale of the camera, light, or other object.

Copying a Viewer's POV to a Camera

There are many times you will have used the default perspective view to set up your scene and then want that point of view (POV) to become your main camera's point of view. You can easily position and orient a camera, light, or other object to match the current point of view shown in the viewer using the Copy PoV To command.

To copy the point of view in the viewer to a camera, do the following:

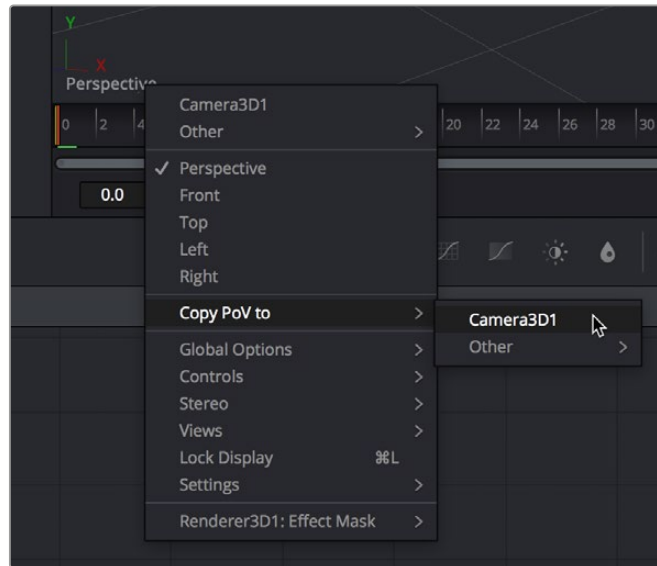
- 1 Set up a 3D Viewer with the point of view you want by zooming, panning, and rotating the viewer.
- 2 Add a camera to your 3D scene.
- 3 Right-click anywhere within the 3D Viewer and choose Camera > Copy PoV To > Camera3DNameOfCamera from the contextual menu.

The Camera3D's controls will inherit the viewer's position and angle values.

TIP: The Copy PoV To command uses the object's own coordinate space; any transformations performed downstream by another node are not taken into account.

POV Labels

As you switch the POV of the viewer, you can keep track of which POV is currently displayed via a text label at the bottom-left corner of the viewer. Right-clicking directly on this label, or on the axis control above it, acts as a shortcut to the Camera submenu, allowing you to easily choose another viewpoint.



The Viewpoint label and Axis control in a 3D Viewer

Lighting and Shadows in 3D Viewers

Before you add lights to a 3D scene, default lighting is provided. This basic, flat lighting allows you to see the shading on objects without requiring you to add and set up lights as you work in the 3D Viewer. Additionally, shadows are hidden by default. Once you start adding lights of your own, you need to switch modes to see what they affect as you work.

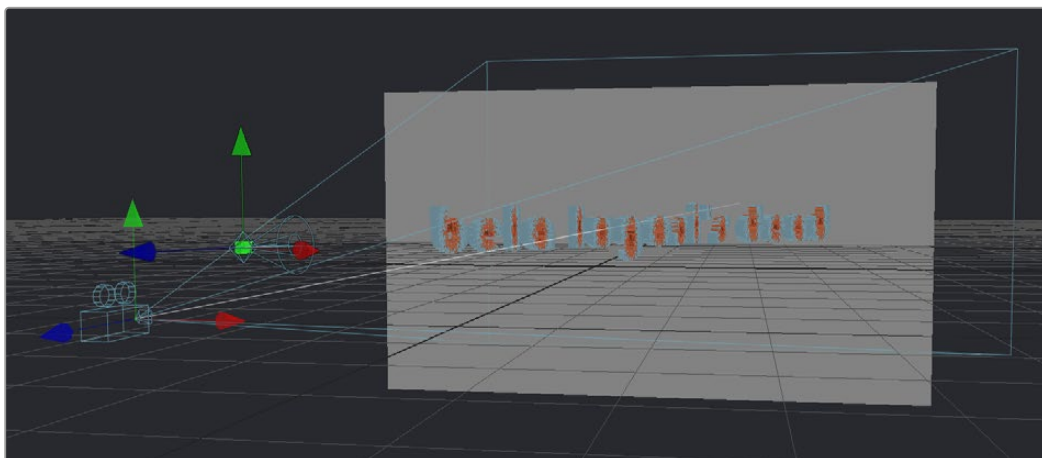
To see the effects of the default light on the scene:

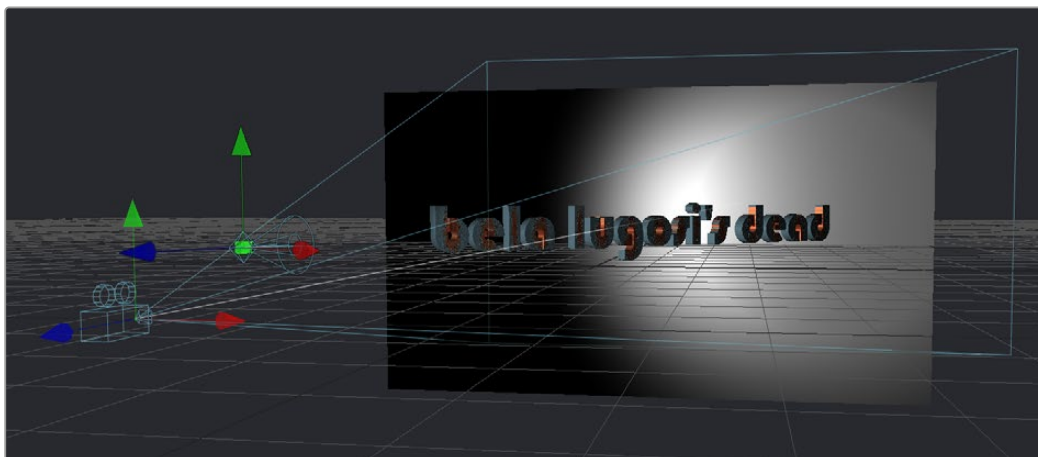
- Right-click within the 3D Viewer and choose 3D Options > Default Lights from the contextual menu.

When you're ready to add your own lighting to a scene, you can connect light nodes in various ways to a Merge 3D node for the scene you're working on. Once you connect a light to a Merge 3D node, you need to switch the 3D Viewer over to showing the new, proper lighting.

To toggle lighting rendering within a 3D scene:

- Right-click within the 3D Viewer and choose 3D Options > Lighting from the contextual menu.





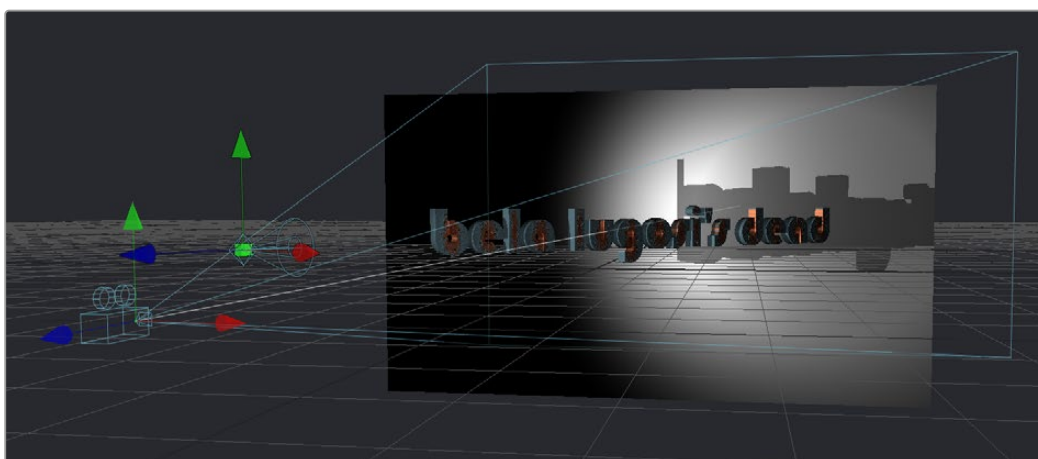
A 3D scene using default lights (top), and the same scene with lighting turned on (bottom)

TIP: Attempting to load a Light node into a viewer all by itself will result in an empty scene, with nothing illuminated. To see the effects of lights, you must view the Merge 3D node the light is connected to.

Similar to lights, the default 3D Viewer has shadows turned off. To see shadows cast from the lighting you've created, you must turn them on.

To toggle shadows rendering within a 3D scene:

- Right-click within the 3D Viewer and choose 3D Options > Shadows from the contextual menu. Enabling shadows will automatically turn on lighting, if it is not already turned on.



A 3D scene with shadows enabled along with the lights

Transparency in 3D Viewers

Image planes and 3D objects are obscured by other objects in a scene depending on the X, Y, and Z position coordinates of each object in 3D space. The default method used to determine which polygons are hidden and which are shown based on these coordinates is called Z-buffering.

Z-buffering is extremely fast but not always accurate when dealing with multiple transparent layers in a scene. Fortunately, there is another option for more complex 3D scenes with transparency: Sorted.

The Sorted method can be significantly slower in some scenes but will provide more accurate results no matter how many layers of transparency happen to be in a scene.

The default behavior in the viewer is to use Z-buffering, but if your scene requires the Sorted method, you can easily change this.

To choose a Sorted method of 3D compositing:

- Right-click anywhere within the 3D Viewer and choose one of the options in the Transparency submenu of the contextual menu;

Full Sort: Renders every polygon in Z order to produce the most accurate rendering of transparency.

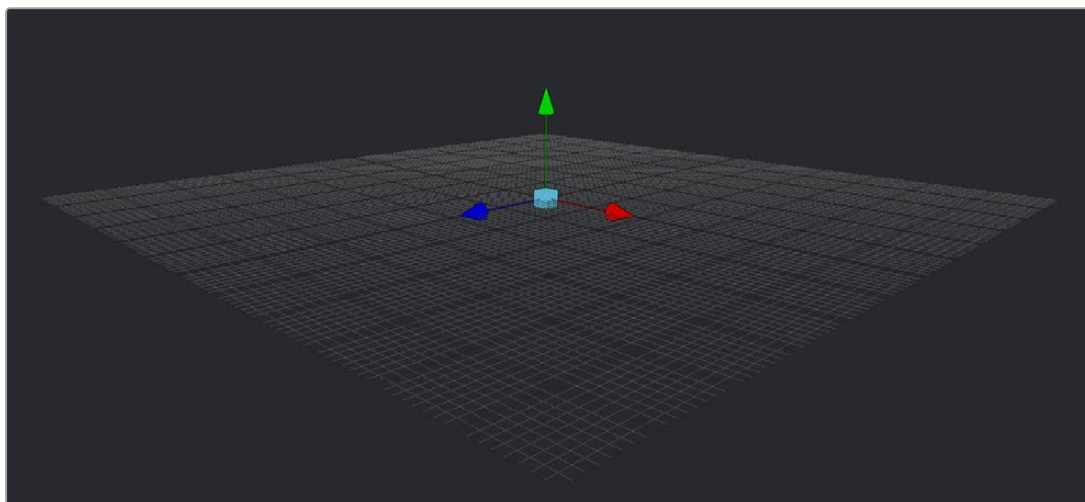
Quick Sort: Reorders the polygons in the scene serially, from back to front, to produce a reasonably accurate rendering of transparency.

Grid

The 3D Viewer displays a grid that's used to provide a plane of reference in the 3D scene. By default, the grid is 24 x 24 units in size, centered on the origin at (0,0,0), and subdivided into large squares of 2 units with small squares of 0.25 units each. These defaults can be altered in the 3D View panel of the Fusion Settings window, available from the Fusion menu.

To toggle the grid on and off:

- Right-click anywhere within the 3D Viewer and choose 3D Options > Grid from the contextual menu.



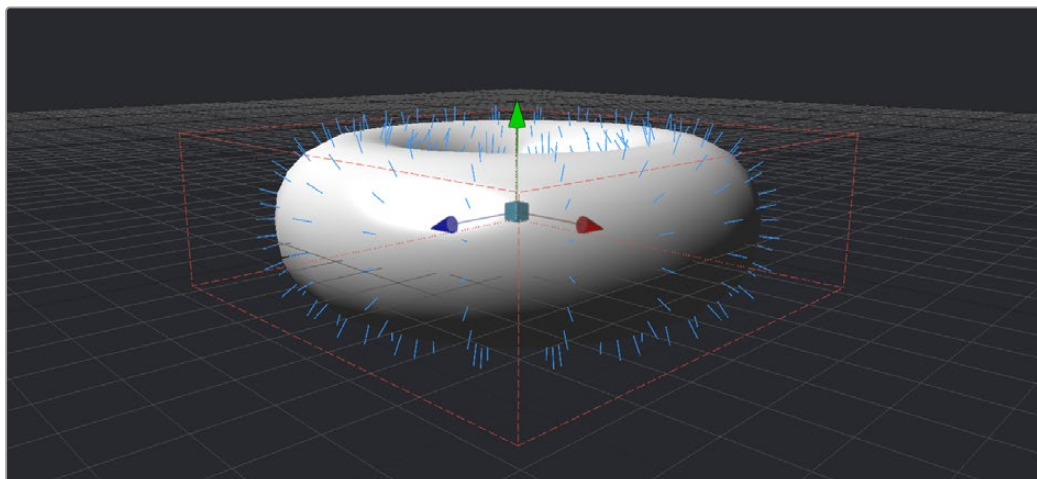
The default grid of the 3D Viewer grid with its origin at $x = 0$, $y = 0$ and $z = 0$

Vertex Normals

Normals indicate what direction each vertex of 3D geometry is facing, and they are used when calculating lighting and texturing on an object. When viewing any kind of 3D geometry, including an image plane or a full FBX mesh, you can display the normals for each object in a scene.

To view the normals in a scene:

- Right-click anywhere within the viewer and choose 3D Options > Vertex Normals from the contextual menu.



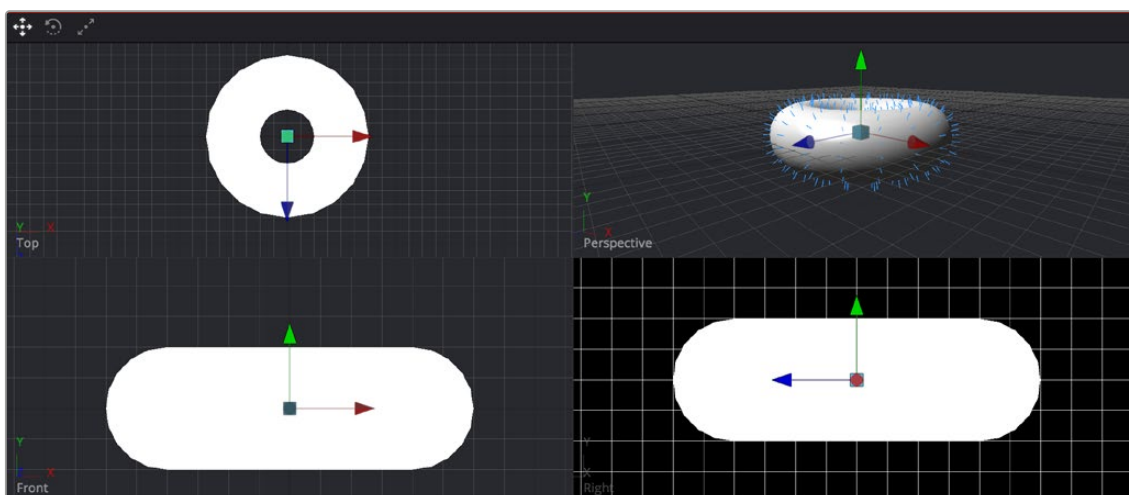
The normals viewed in a 3D scene

Quad View

3D compositing often requires you to view the scene from different points of view to better control transformations in three dimensions. While you can switch the 3D Viewer to different points of view, doing so frequently can become cumbersome. Happily, you can instead enable a Quad view, which divides the viewer into four panes. These panes can then display four different angles of the scene at one time.

To toggle the display of the Quad view, do one of the following:

- Right-click anywhere within the viewer and choose Views > Quad View from the contextual menu.
- Press Shift-Q.



A Quad view of a 3D scene

While there are four panes in the Quad view, they all show the same scene. When assigning views within a Quad view, you can choose between displaying Front, Left, Top, Bottom, and Perspective orthographic views, or you can choose the view through any camera or spotlight that's present in the scene.

To assign different views to panes of a Quad view, do one of the following:

- Right-click directly on the POV label at the bottom left of the pane you want to reassign, and choose another camera, light, or Point of View from the contextual menu.

Quad View Layouts

There are a variety of Quad view layouts, ranging from four equally sized panels to having three small panels across the bottom of a larger single panel.

To switch to a different Quad view layout, do the following:

- 1 Enable the Quad view.
- 2 Right-click anywhere within the viewer and choose an option from the Views > Quad Layouts submenu of the contextual menu.

Using Quad Views for 2D Scenes

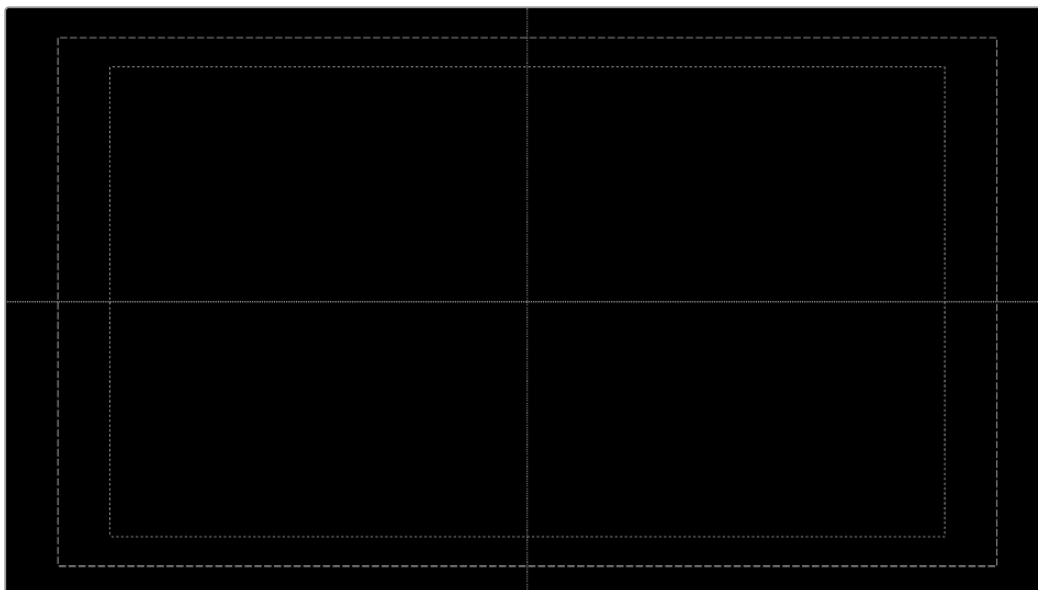
Quad views aren't only useful for 3D scenes. They can also be used with 2D scenes, with each pane showing a different image channel or subview type. For example, one pane can show the image while the other panes show the alpha channel, a vectorscope, and a histogram.

To assign different channels or subviews to panes of a Quad view for a 2D scene:

- 1 Right-click in a viewer and choose Views > Quad View.
- 2 Click once in the pane you want to reassign.
- 3 Do one of the following:
 - a) Choose a channel from the Channel Viewer menu.
 - b) Right-click in the viewer and choose Views, and then choose a Subview from the submenu.

Guides

Guides are onscreen overlays used to help you compose elements within a boundary or along the center vertical and horizontal axes. While guides are displayed in the viewer, they're not rendered into the scene. There are four commonly used guides that can be displayed, including Monitor Safety, Safe Title, Center, and Film.



The Guides submenu in the viewer's contextual menu

Methods of using guides:

- **To display guides in a viewer:** Right-click in the viewer and then choose Guides > Show Guides from the contextual menu, or press Command-G.
- **To change the aspect ratio of the displayed guides:** Right-click in the viewer and then choose an option from the Guides > Frame Aspect submenu. The frame aspect is usually set to Default, which forces the frame aspect to the same resolution as the image that's displayed in the view. However, when the frame aspect is set to a specific value, the guides will conform to the exact boundaries of the specified format and any image area outside of that will be dark gray.
- **To show or hide specific guides:** Right-click in the viewer and then choose an option from the Guides submenu. A variety of specific guides are provided, each of which can be individually enabled and disabled.

Monitor Safety: Monitor Safety indicates the safe action area viewable on most monitors and TV screens.

Safe Title: Safe Title indicates the safe area for titles viewable on all TV and monitor screens.

Center: Center shows a crosshair for the center point and x- and y-axis of the view.

Film: Some frame formats include film guides preset for you, whereas some will require customization. The film guides can be customized in the Preferences > Frame Format window.

Frame Format Settings

In the Frame Format panel of the Fusion Settings window (available in the Fusion menu), there are two film guide settings that you can use to customize these guides.

- Guide 1 contains four fields that specify the offset from the edges of the image for the left, top, right, and bottom guides, in that order. As with all offsets in Fusion, this is a resolution-independent number where 1 is the width of the full image and 0.5 is half the width of the image.
- Guide 2's text box is used to set the aspect ratio of the projection area.



The Frame Format Guides settings

Domain of Definition and Region of Interest

As a compositing environment, the Fusion page uses the standard compositing conventions of Region of Interest (RoI) and Domain of Definition (DoD) to dramatically improve performance.

Domain of Definition (DoD)

In compositing, the Domain of Definition, frequently abbreviated to DoD, refers to a rectangular region that defines what part of an image actually contains data. DoD makes the concept of an image's actual frame somewhat flexible, since rendering is no longer limited to the actual width and height of the image. This has two effects on the way Fusion renders images.

Firstly, nodes will no longer be required to render portions of the image that will not be affected by the node. This helps the renderer to optimize its performance. Secondly, Fusion can now keep track of and apply a node's effect to pixels that lie outside the visible portion of the image.

For example, consider the output of a Text+ node rendered against a transparent background. The text occupies only a portion of the pixels in the image. Without Domain of Definition, you would be required to process every pixel in the image needlessly. With a DoD, you are able to optimize effects applied to the image, producing faster results and consuming less memory in the process.

The following image shows an image with the DoD outlined.



The DoD is shown as two XY coordinates indicating the corners of an axis-aligned bounding box (in pixels)

For the most part, the DoD is calculated automatically and without the need for manual intervention. For example, all the nodes in the Generator category automatically generate the correct DoD. For nodes like Fast Noise, Mandelbrot, and Background, this is usually the full dimensions of the image. In the case of Text+ and virtually all of the Mask nodes, the DoD will often be much smaller or larger.

The OpenEXR format is capable of storing the data window of the image, and Fusion will apply this as the DoD when loading such an image through a Loader node and will write out the DoD through the Saver node.

When using the Fusion page in DaVinci Resolve, clips from the Edit page timeline or Media Pool will typically have the DoD default to the full image width of the source media. The exception is media stored in OpenEXR format.

The DoD is established as soon as the image is created or loaded into the composition. From there, it passes downstream, where viewers combine it with their Region of Interest in order to determine exactly what pixels should be affected by the node. As you work, different nodes will automatically shrink, expand, or move the DoD as they apply their effect to an image, causing the DoD to change from node to node.

Showing the DoD

If the current DoD for a node is different from the frame size of that image, it's shown in the tooltip that appears when the pointer hovers over a node in the Node Editor. The DoD is also visible in the viewer when you right-click in a viewer and choose Region > Show DoD from the contextual menu.

Setting the DoD Manually in the Node Editor

It is also possible to set the DoD for an image manually using the Tools > Miscellaneous > Auto Domain node in the Effects Library. This node can be useful when dealing with pre-created media that does not occupy the full image dimensions. For example, a rendering of a 3D character that walks toward the camera will frequently occupy only a portion of the image. The Auto Domain node can be used to animate a DoD that covers the character and ignores the rest of the image, making image processing more efficient.

Region of Interest (RoI)

The Region of Interest, frequently abbreviated to RoI, is a rectangular region similar to the Domain of Definition. However, unlike the DoD, which tells the node what pixels are actually present in the image, the RoI tells the node which pixels actually need to be rendered. When a node renders, it intersects the current RoI with the current DoD to determine what pixels should be affected.

Enabling RoI Controls

You can turn on the RoI controls to restrict rendering to a small region of the image to significantly improve performance when you're only working on a small part of a high-resolution or complex composition. For example, if you're using paint to clean up some holes in a matte on the floor of a composition with many, many high-resolution layers, 3D, and Lighting operations, you can use the RoI controls to isolate the part of the floor you're working on, which makes caching that part of the composition much faster.

To enable the RoI controls, do one of the following:

- Click the RoI button in the 2D Viewer toolbar.
- Right-click in a viewer and choose Region > Show Region from the contextual menu.

When RoI is enabled and Show Region is selected from the menu, a rectangular RoI control appears in the viewer. If this is the first time RoI has been enabled, it will be set to the full width and height of the image. Otherwise, the last known position of the RoI for that view is used. However, if you want to set the RoI to a custom area within the frame, you can do one of the following.

To adjust the RoI controls, do one of the following:

- Drag any edge of the RoI rectangle to adjust one side of the RoI.
- Drag a corner to adjust the size of the RoI rectangle from that corner.
- Drag the small circle found at the top left corner of the RoI rectangle to move the RoI without adjusting its dimensions.

Sometimes, it's faster to simply draw a rectangle where you want the RoI to be.

To quickly draw the RoI at the desired size:

- 1 Choose Set from the viewer menu next to the RoI button, or right-click anywhere within the viewer and choose Region > Set Region.
- 2 When the pointer turns into an RoI drawing cursor, drag within the viewer to set a RoI rectangle.

Alternatively, an Auto command sets the RoI to fit whichever pixels are visible at the current zoom/pan level in the viewer. This lets you quickly limit the RoI to whatever part of the composition you've zoomed into.

To automatically draw the RoI:

- Choose Auto from the viewer menu next to the RoI button.
- Right-click anywhere within the viewer and choose Region > Auto Region.
- When you no longer need to use the RoI, you can reset it.

To reset the RoI to the full width and height of the current image, do one of the following:

- Choose Reset from the viewer menu next to the RoI button.
- Right-click anywhere within the viewer and choose Region > Reset Region from the contextual menu or from the toolbar button menu.
- Disable the ROI control, which will also reset it.

While the RoI Is Active

The RoI is only used for previewing your composition while you work, not for output from Fusion. While the RoI is active, Fusion will only request rendering of the pixels inside the region when it displays an image in that viewer. Flipbook Previews that you create in that viewer will also respect the current RoI. MediaOut and Saver nodes will always use the full image dimensions when writing the image to disk, ignoring any RoI you've set in the viewers.

The RoI improves not only rendering speed and memory use, but it can also reduce file I/O, since Loaders and MediaIn nodes only load pixels from within the RoI, if one is specified. This does require that the file format used supports direct pixel access. Cineon, DPX, and many uncompressed file formats support this feature, as well as OpenEXR and TIFF in limited cases.

Please note that changes to the viewed image size or color depth will cause the pixels outside the RoI to be reset to the image's canvas color. This also happens when switching in and out of Proxy mode, as well as during Proxy mode switching with Auto Proxy enabled. When the image size is maintained, so are the last rendered pixel values outside the RoI. This can be useful for comparing changes made within the RoI with a previous node state.

TIP: Right-clicking in a viewer and choosing Options > Show Controls for showing onscreen controls will override the RoI, forcing renders of pixels for the entire image.

Managing Viewer Lookup Tables (LUTs)

Lookup Tables, or LUTs, can be used to help match the appearance of a viewer to its eventual output destination. They're essentially image-processing operations that affect only the image being previewed in the viewer, not the image data itself. There are two basic ways that LUTs can calculate color transformations: The first is a simple 1D LUT, and the second is a more sophisticated 3D LUT.

- The simplest form of a LUT is a 1D LUT. It accounts for one color channel at a time, so it can make overall tonality changes but not very specific color changes.
- A 3D LUT looks at each possible color value (red, green, and blue) independently. A 3D LUT allows for large global changes as well as very specific color changes to be applied to images very quickly.

How Lookup Tables Work in Fusion

A Lookup Table (LUT) is a table of values used to transform the color and luminance of an image. A 1D LUT uses a two-column table for input color and output color, while a 3D LUT uses more of a matrix. A LUT is used primarily to correct for variances in the monitor or the source color space of the image. You can choose to apply a LUT to all the viewers or apply different LUTs to each viewer.

Image LUTs

Image LUTs can be applied to each viewer. In fact, you can even apply separate Image LUTs for the A and B buffers of a single viewer. These LUTs can only be applied to 2D images and not to 3D scenes. Image LUTs are routinely used to get from one scene-referred color space to another. For example, if you're working with log-encoded media but want to see how the image will look in the final color space, you can choose a LUT to make the image transform as a preview.

Buffer LUTs

The Buffer LUT is applied to the viewers regardless of contents, including 3D scenes, 3D materials, and subview types. Only one Buffer LUT can be applied. If a 2D image is being displayed with an Image LUT applied, then the Buffer LUT is applied to the result of the image LUT. Buffer LUTs are typically used to simulate another output color space that's unique to the display you're using—for instance, making a DCI-P3 projector show the image as it would look on an sRGB monitor.

To use a Buffer LUT:

- 1 Disable the LUT button above the viewer.
- 2 Right-click in the viewer and choose Global Options > Buffer LUT > Enable.
- 3 Right-click in the viewer and choose Global Options > Buffer LUT > Type of LUT you want to apply.

When dealing with nonlinear files from many of today's digital cinema cameras, a modern workflow would be to convert everything to linear at the beginning of the node tree, then create your composite, and then apply an Image LUT or Buffer LUT that matches the color space you want it to be in for either grading in the Color page or for final output.

However, in more elaborate production pipelines, you may need to apply multiple LUTs consecutively.

Types of Viewer LUTs

Aside from the industry standard 1D and 3D LUTs, other types of LUTs are supported, including script-based Fuse node LUTs, OCIO files, and macros assembled from standard nodes. Generally, LUT processing is performed on the graphics card's GPU in real time, although the performance of macro-based LUTs is based on the nodes they contain.

Fusion View LUT

The Fusion View LUT is the default and is a frequently used LUT type. It provides an RGBA curve that can be used to assign IN/OUT value pairs. This control is identical to that provided by the Color Curve node.

Since the purpose of the View LUT is to provide an unchanging correction for the monitor or the file's color space, however, these splines cannot be animated.

Log-Lin View LUT

The Log-Lin LUT converts logarithmic data to linear, and vice versa. This can be particularly useful when used in conjunction with supplied LUT files that expect logarithmic data. It is similar to the Cineon Log node.

Gamut View LUT

The Gamut LUT converts a source color space to an output color space, with options to deal with gamma settings, alpha channels, and premultiplication. The Gamut LUT is a frequently used LUT type to correct the viewer when working with Linear Gamma in the Node editor.

Macro LUTs

Any macro node can also be used as a viewer LUT simply by saving the macro's .setting file to the correct Fusion directory.

In DaVinci Resolve, LUTs are saved in the following locations:

On macOS: Macintosh HD/Users/username/Library/Application Support/Blackmagic Design/Fusion/LUTs/

On Windows: C:\Program Files\Blackmagic Design\Fusion\LUTs

On Linux: home/username/.local/share/DaVinciResolve/Fusion/LUTs

In Fusion Studio, LUTs are saved in the following locations:

On macOS: Macintosh HD/Users/username/Library/Application Support/Blackmagic Design/Fusion/LUTs/

On Windows: C:\Users\username\AppData\Roaming\Blackmagic Design\Fusion\LUTs

On Linux: home/username/.fusion/BlackmagicDesign/Fusion/LUTs

For this to work, the macro must have one image input and one image output. Any controls exposed on the macro will be available when the Edit option is selected for the LUT.

For more information about creating macros, see *Chapter 68, "Node Groups, Macros, and Fusion Templates,"* in the DaVinci Resolve Reference Manual or Chapter 6 in the Fusion Reference Manual.

LUT Presets

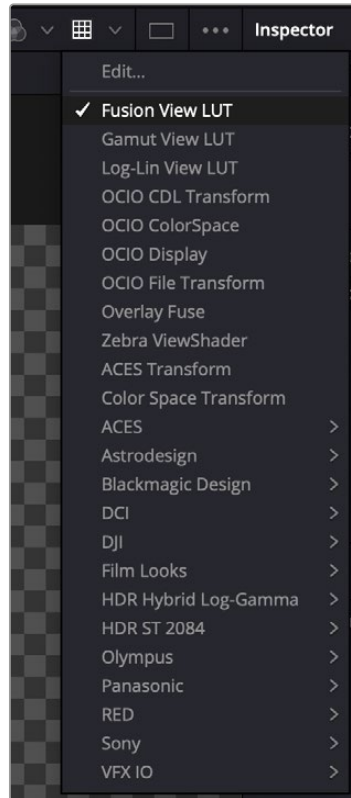
All LUTs available to DaVinci Resolve are also accessible to the Fusion page, which includes custom LUTs you've installed, as well as preset LUTs that come installed with DaVinci Resolve, such as the highly useful VFX IO category that includes a wide variety of miscellaneous to Linear and Linear to miscellaneous transforms. All of these LUTs appear by category in the viewer LUT menu.

Fuse LUTs

Fuses are scriptable plugins that are installed with the application or that you create in Fusion. A fuse named CT_ViewLUTPlugin can be applied as a LUT to a viewer. You can also script fuses that use graphics hardware shaders embedded into the LUT for real-time processing. Since fuse LUTs require shader-capable graphics hardware, they cannot be applied in software. For more information about Fuses, see the Fusion Scripting Guide located on the Blackmagic Design website.

Using Viewer LUTs

Viewer LUTs can be enabled, edited, and turned off using the viewer LUT button and menu, as well as by using the viewer contextual menu. This menu shows all LUTs available to Fusion, including custom LUTs you've installed yourself.



The viewer LUT button and menu from the Fusion page in DaVinci Resolve

To turn the current viewer LUT on and off:

- Click the LUT button in the viewer toolbar to toggle the viewer LUT on and off.
- The LUT menu can also be found as a submenu in the viewer's contextual menu.

To choose another viewer LUT:

- Open the menu to the right of the viewer LUT button and choose an option from the viewer LUT menu.

To apply a Buffer LUT:

- 1 Right-click anywhere within the viewer and choose Global Options > Buffer LUT > Enable.
- 2 To choose a specific Buffer LUT, right-click again and choose a LUT from the Global Options > Buffer LUT submenu.

Buffer LUTs are often useful for applying monitor corrections, which do not usually change between projects.

To remove a Buffer LUT:

- Right-click anywhere within a viewer and choose Global Options > Buffer LUT > Enable to uncheck it.

Editing Viewer LUTs

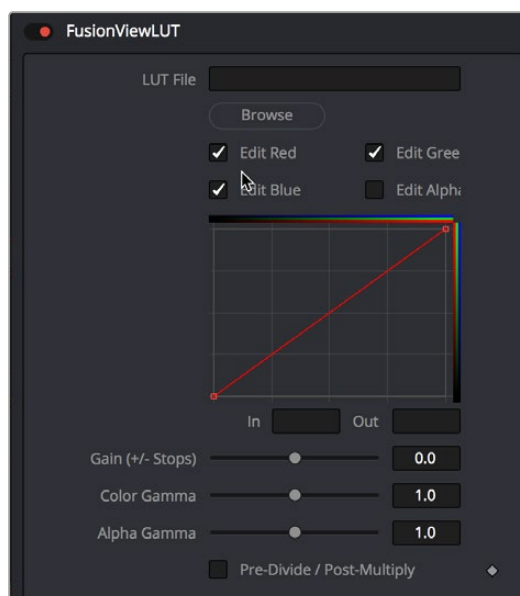
The viewers are the primary area where composites are assessed, so it's crucial that they provide an accurate representation of what the content will look like when it's played for an audience. The LUT Editor allows you to customize your viewer's output to match the gamma and color characteristics of your eventual playback device, or to test how the current image looks in a completely different color space, or how it holds up over a range of different color spaces.

To open any editable viewer LUT option's Editor:

- 1 Click the LUT button in the viewer toolbar to enable it.
- 2 Do one of the following:
 - Choose Edit from the top of the viewer LUT menu.
 - Right-click in the viewer and then choose LUT > Edit from the contextual menu.

Editing the Fusion View Lookup Table

Similarly to the Color Curves node, the Fusion View LUT Editor uses spline-based color correction. In addition to the ability to modify the separate color channels, the LUT has Gain and Gamma sliders. The Gain slider is helpful for temporarily brightening or darkening the viewed image, allowing easier examination of shadow or highlight detail. The Color Gamma and Alpha Gamma sliders are used to duplicate the gamma values of the eventual output device. Video monitors, for example, commonly have a gamma of 1.7, while computer monitors can range anywhere from 1.6 to 2.2. Alpha Gamma is applied only when viewing the alpha channel of an image, or when viewing masks.



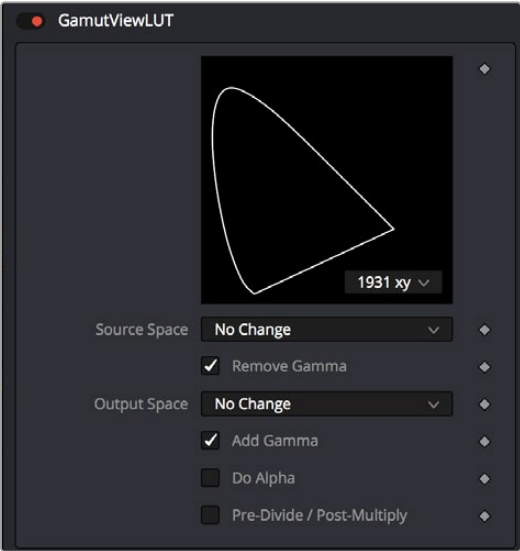
The LUT Editor for the default Fusion View LUT

Editing the Gamut View LUT

The Gamut View LUT Editor lets you choose a Source and Output color space to guide the viewer transform.

The Remove and Add Gamma checkboxes let you choose to do the gamut conversion with linear or nonlinear gamma, or they let you simply remove or add the appropriate gamma values without changing the color space.

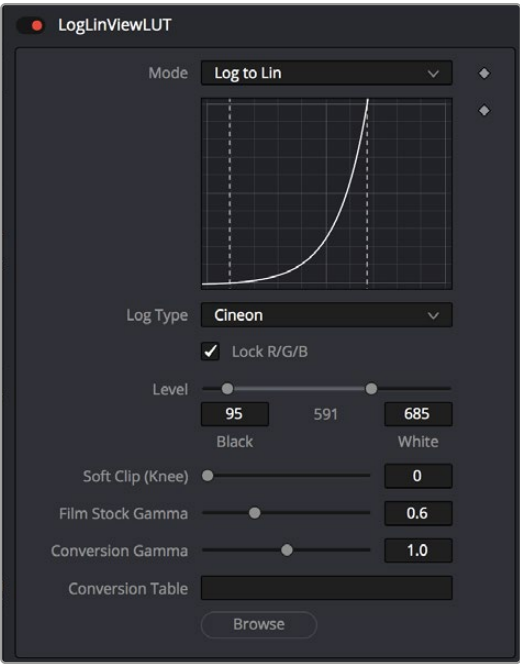
Selecting the Pre-Divide/Post-Multiply checkbox will cause the image's pixel values to be divided by the alpha values prior to this conversion, and then re-multiplied by the alpha value after this conversion. This helps to avoid the creation of illegally additive images, particularly around the edges of a blue/green key or when working with 3D rendered objects.



The Gamut View LUT Editor

Editing the Log-Lin View LUT

The Log-Lin LUT lets you apply a Log to Lin or Lin to Log operation using the Mode pop-up menu. You can choose the type of log-encoding to process from the Log Type drop-down, and choose whether to lock the R, G, and B channels together. A level adjustment lets you redefine the digital range of values used for the output, while Soft Clip (Knee), Film Stock Gamma, and Conversion Gamma sliders let you further customize the color transform. Lastly, a Conversion Table field and Browse button let you add an additional LUT as part of this operation.



The Log-Lin LUT Editor

LUT Processing Order

In elaborate workflows, facilities may apply multiple LUTs in a row before the image is seen. The order of these is important since each LUT delivers different outputs. For instance, for a Cineon file in Log color space you may often apply three LUTs. First a Log to Lin conversion, followed by a Fusion View LUT to apply a color calibration, and a third one to correct it for display on an sRGB monitor, or replace the last with a 3D DCP LUT if you are viewing on a projector.

When you select a node to be displayed, the image produced is processed before it is shown in the viewers. The processing order is slightly different for 2D images and 3D scenes.

2D images first have the image LUT applied, and the result is composited over the checker underlay. 3D scenes are instead rendered with OpenGL.



The order of processing for 2D images and 3D scenes

For either 2D or 3D, the result may be drawn to an offscreen buffer where a Buffer LUT can be applied, along with dithering, a full view checker underlay, and any stereo processing. The final result is then drawn to the viewer and any onscreen controls are drawn on top.

Applying Multiple LUTs

The viewer contextual menu can be used to apply multiple image LUTs into a processing chain.

To apply an additional LUT, do the following:

- 1 Right-click anywhere within the viewer.
- 2 From the viewer's contextual menu, choose LUT > Add New.
- 3 From the Add New submenu, choose a LUT to add.

To remove a LUT other than the first LUT, do the following:

- 1 Right-click anywhere within the viewer.
- 2 From the viewer's contextual menu, choose LUT > Delete.
- 3 From the Delete submenu, choose a LUT to remove.

A complete stacked LUT configuration can be saved to and loaded from a .viewlut file, as described below.

Saving Custom LUTs

There are a variety of ways to create and use different viewer LUTs in Fusion. You can save LUTs when you save viewer settings, you can import LUTs that have been exported from Fusion or other applications, and you can open any one of the various supported LUT file types. In addition, you can use the standard nodes in Fusion to create macros, which can then be saved and used as a LUT.

LUT Settings

The most straightforward way to save a LUT you have created using the Fusion View LUT Editor is to use the LUT > Save menu found in the viewer contextual menu. The settings are saved as an ASCII file with the extension .viewlut in the LUTs folder. Any files with this extension found in that folder will appear in the Image LUT menus for ease of loading. You can also load the settings that are not found in the menu by choosing LUT > Load from the viewer's contextual menu.

Using Viewer Settings

If you've modified a LUT, choosing Settings > Save New from the viewer's contextual menu will save all the viewer's settings, including all LUT curves and gain/gamma values. You can save these under different names, and each settings file can be reloaded at any time by choosing Settings > filename from the viewer's contextual menu. Choosing Save Default from the same menu will make these settings the standard for all new comps.

Using LUT Curves

The Viewer LUT Edit dialog can be used to import and export LUT curves. You can export the LUT curves as either ASCII or Saved format. The ASCII (.alut) file format is useful for sharing LUT curves with other software, whereas the Saved (.lut) file format is preferred for Fusion, as it is more compact, accurate, and allows further editing.

To export a LUT, do the following:

- 1 Click the viewer LUT button to enable it.
- 2 Click the viewer LUT menu, and then choose Edit.
- 3 Right-click on the LUT Curve Editor, and then choose Export LUT.
- 4 Select a LUT format at the bottom of the file browser window.
- 5 Enter a name for the LUT and click Save.

The Import LUT option will load LUT files back into the Curve Editor, or alternatively, if the file has been saved in Fusion's LUTs folder, it will appear in the LUT drop-down menu list.

TIP: This is one way to move LUTs between viewers or to and from the Color Curves node or any other LUT Editor in Fusion.

LUT Files

Any supported LUT files in the LUTs folder can be used by choosing them either from the LUT drop-down menu or the viewer's contextual menu. This includes 1D and 3D LUTs such as Fusion's .lut, .alut and .alut3 formats, as well as .cube, .shlut, .look, .3dl, and .itx formats. This is a convenient way to access standard format LUT files for different projects.

Settings and Macros

Since LUTs are a form of color correction, you can also use any node, macro, or group of nodes as a viewer LUT.

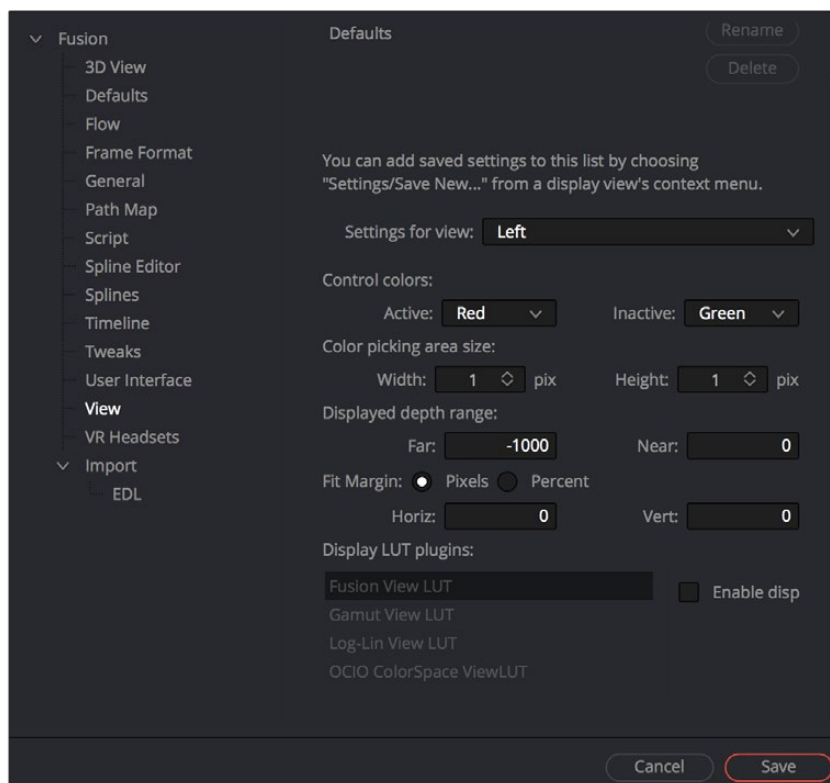
To use a node, group, or macro as a viewer LUT, do the following:

- 1 Select the node, group, or macro.
- 2 Right-click over the selected node, and then choose Settings > Save As from the menu.
- 3 In the file browser, go to the LUTs folder as set in Preferences > Global > Path Map > LUTS.
- 4 Click Save to save the .settings file.

This allows almost any combination of nodes to be used as a viewer LUT. This is the most flexible approach but is also potentially the slowest. The LUT nodes must be rendered solely on the CPU, whereas other methods are GPU-accelerated.

Setting a Default LUT

The default LUT applied when a new composition is created can be assigned in the Viewer panel of the Fusion Settings window. Clicking the Enable Display LUT checkbox allows you to select a LUT from the Display LUT plugins list.



The LUT default settings found in the View panel of the Fusion Settings window

Viewer Preferences and Settings

The default settings for each viewer can be changed using the Viewer panel in the Preferences. The position and size of each floating viewer can also be saved using the Layout menu in the Preferences.

Viewer Settings

It is often preferable to switch between entirely different viewer configurations while working. For example, while keying, the image may be in the main viewer, and the alpha channel may be in a subview. Viewer settings toward the end of a project may consist of the histogram, vectorscope, and waveform, as well as the image in a view set to Quad view.

Fusion provides the ability to quickly load and save viewer settings to help reduce the amount of effort required to change from one configuration to another.

To save a viewer setting, do the following:

- 1 Right-click over the viewer you want to save.
- 2 From the contextual menu, choose Setting > Save New.
- 3 Enter a name for the settings and click Save.

To load a viewer setting, do the following:

- 1 Right-click over the viewer you want to load a setting into.
- 2 From the contextual menu, choose Settings > filename.

Loading and Saving Defaults for a Viewer

The viewer can save new defaults and be returned to its defaults using the Load Defaults and the Save Defaults options in the Settings portion of the View contextual menu.

The Viewer Options Menu

The Options menu of the viewer contains several ways you can customize the look and behavior of the viewer. Many of these options are also in the viewer contextual menu.

Show Controls

When onscreen controls are not necessary or are getting in the way of evaluating the image, you can temporarily hide them using the Show Controls option. This option is toggled using Command-K.

Checker Underlay

The Checker Underlay shows a checkerboard beneath transparent pixels to make it easier to identify transparent areas. This is the default option for 2D viewers. Disabling this option replaces the checkerboard with black.

Show Pixel Grid

Enabling this option will show a light black grid that outlines the exact boundaries of pixels in the image when the image is scaled past a certain threshold. The default is Off.

Smooth Resize

The Smooth Resize option uses a smoother bilinear interpolated resizing method when zooming into an image in the viewer. When Smooth Resize is disabled, scaling uses the nearest neighbor method and shows noticeable aliasing artifacts but is more useful for seeing the actual pixels of the viewed image when you zoom all the way down to a pixel level since there is no interpolation. This option is enabled by default and can be toggled by clicking on the SmR button in the viewer toolbar.

Show Square Pixels

Depending on the frame format preferences and the type of footage loaded, many images may have pixels that are rectangular instead of square. Both the NTSC and PAL video standards, as well as some anamorphic film formats, use rectangular pixels. A computer monitor uses perfectly square pixels. To compensate for this, aspect correction is automatically performed when viewing non-square pixels. This prevents non-square pixel images from appearing squashed or stretched in the viewer.

You can enable the Show Square Pixels option to override the aspect correction. Show Square Pixels can also be toggled on and off using the 1:1 button in the viewer toolbar.

Gain/Gamma

Exposes or hides a simple pair of Gain and Gamma sliders that let you adjust the viewed image. Especially useful for “gamma slamming” a composite to see how well it holds up with a variety of gamma settings. Defaults to no change.

360° View

Sets the Fusion page viewer to properly display spherical imagery in a variety of formats, selectable from this submenu. Disable toggles 360 viewing on or off, while Auto, LatLong, Vert Cross, Horiz Cross, Vert Strip, and Horiz Strip let you properly display different formats of 360° video.

Locking the Viewer (Command-L)

You can lock a viewer to prevent it from updating. The node that’s loaded into that viewer still processes and the new image is queued for display in the viewer, but until you unlock it, the viewer does not update. By default, the viewer is unlocked.

Additional Viewer Options

There are additional commands when you right-click anywhere within a viewer and choose from the generically named Options submenu.

Alpha Overlay

When you enable the alpha overlay, the viewer will show the alpha channel overlaid on top of the color channels. This can be helpful when trying to see where one image stops and another begins in a composite. This option is disabled by default.

Overlay Color

When you turn the alpha overlay on, the default color is to show white for the area the alpha covers. There are times when white does not show clearly enough, depending on the colors in the image. You can change the color by choosing a color from the list of Overlay Color options.

Follow Active

Enabling the Follow Active option will cause the viewer to always display the currently active node in the Node Editor. This option is disabled by default, so you can view a different node than what you control in the Control Panel.

Show Controls

When onscreen controls are not necessary or are getting in the way of evaluating the image, you can temporarily hide them using the Show Controls option. This option is toggled using Command-K.

Show Full Color Range

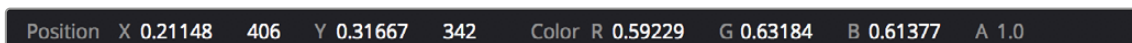
When working with floating-point images, you will occasionally need to visualize the values that fall outside the normal luminance range. Enabling the Show Full Color Range option using the toolbar button automatically normalize any image displayed in the viewer. Normalization causes the brightest pixel in a color channel to be mapped to a value of 1.0 (white) and the darkest pixel to be mapped to a value of 0.0 (black). Midrange values are scaled appropriately to fit within that range. It is also useful when viewing Z-buffer or other auxiliary channels, which often use value ranges far different from those in the color channels.

Show Labels

The Show Labels option lets you toggle the display of the text that sometimes accompanies onscreen controls in the viewer without disabling the functions that are showing those overlays, and without hiding the onscreen controls themselves.

Status Bar Information

The status bar at the bottom of the Fusion window provides the exact RGBA and Z values for the pixel beneath the pointer when it's hovering within one of the viewers. Additional information about the X and Y coordinates of the cursor and the exact pixel position are also displayed.



The status bar displays the following information: Position X 0.21148 406 Y 0.31667 342 Color R 0.59229 G 0.63184 B 0.61377 A 1.0

The status bar showing coordinates and color information